

BINOMIAL THEOREM



SYNOPSIS

BINOMIAL THEOREM FOR POSITIVE INTEGER INDEX

GENERAL TERM OF $(x + a)^n$ AND

COEFFICIENT OF x^n

- If n is positive integer, then $(x + a)^n = \sum_{r=0}^n {}^nC_r x^{n-r} a^r$
 $= {}^nC_0 x^n + {}^nC_1 x^{n-1} a + {}^nC_2 x^{n-2} a^2 + \dots + {}^nC_n a^n$
- i) The number of terms in the expansion $(x + a)^n$ is $(n + 1)$
- ii) The sum of the powers of x and a in any term in the expansion of $(x + a)^n$ is n .
- iii) The general term in the expansion of $(x + a)^n$ is $T_{r+1} = {}^nC_r x^{n-r} a^r$
- iv) $(1 + x)^n = \sum_{r=0}^n {}^nC_r x^r$
 $= {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \dots + {}^nC_n x^n$
- v) $T_{r+1} = {}^nC_r x^r$
- vi) ${}^nC_0, {}^nC_1, {}^nC_2, \dots, {}^nC_n$ are binomial coefficients in the expansion of $(x + a)^n$
- vii) The binomial coefficients which are equidistant from the beginning and from the ending are equal.
 i.e. ${}^nC_0 = {}^nC_n; {}^nC_1 = {}^nC_{n-1}; {}^nC_2 = {}^nC_{n-2}; \dots$ etc
- viii) The expansions of $(x + a)^n$ and $(a + x)^n$ are equal but their respective terms are not equal.

W.E-1: The coefficient of x^9 in $(1 + 3x + 3x^2 + x^3)^{15}$ is

Sol. $\left((1 + x)^3\right)^{15} = (1 + x)^{45}$

$$T_{r+1} = {}^{45}C_r x^r; \text{Coefficient of } x^9 = \frac{45!}{36!9!}$$

W.E-2: The coefficient of x^p in the expansion of

$$\left(x^2 + \frac{1}{x}\right)^{2n}$$

Sol. General term $T_{r+1} = {}^{2n}C_r (x^2)^{2n-r} \left(\frac{1}{x}\right)^r = {}^{2n}C_r x^{4n-3r}$

On putting $4n - 3r = p \Rightarrow r = \frac{4n - p}{3}$

coefficient of x^p is $\frac{(2n)!}{\left(\frac{4n-p}{3}\right)! \left(\frac{2n+p}{3}\right)!}$

W.E-3: The coefficient of x^{-n} in $(1 + x)^n \left(1 + \frac{1}{x}\right)^n$ is

Sol. $= \frac{(1 + x)^{2n}}{x^n}$

$\therefore$ coefficient of x^{-n} is coefficient of x^0 in $(1 + x)^{2n}$ is ${}^{2n}C_0 = 1$

➤ $(x - a)^n = \sum_{r=0}^n (-1)^r {}^nC_r x^{n-r} a^r$
 $= {}^nC_0 x^n - {}^nC_1 x^{n-1} a + {}^nC_2 x^{n-2} a^2 - \dots + (-1)^n {}^nC_n a^n$
 The general term of this expansion is

$$T_{r+1} = (-1)^r {}^nC_r x^{n-r} a^r$$

W.E-4: If $ab \neq 0$ and the sum of the coefficients

of x^7 and x^4 in the expansion of $\left(\frac{x^2}{a} - \frac{b}{x}\right)^{11}$ is zero, then [EAM-2012]

Sol. $T_{r+1} = {}^{11}C_r \left(\frac{x^2}{a}\right)^{11-r} \left(\frac{-b}{x}\right)^r$

$$x^{22-3r} = x^7, x^{22-3r} = x^4$$

$$22 - 3r = 7, 22 - 3r = 4; r = 5, r = 6$$

coefficient of $x^7 = -{}^{11}C_5 \left(\frac{b^5}{a^6}\right); x^4 = {}^{11}C_6 \left(\frac{b^6}{a^5}\right)$

$${}^{11}C_6 \left(\frac{b^6}{a^5}\right) - {}^{11}C_5 \left(\frac{b^5}{a^6}\right) = 0; b - \frac{1}{a} = 0, ab = 1$$

W.E-5: If the coefficients of x^3 and x^4 in the expansion of $(1+ax+bx^2)(1-2x)^{18}$ in powers of x are both zero, then (a, b) is equal to [MAINS-14]

Sol. Coefficient of $x^3 = \text{coeff. of } x^3 \text{ in } (1-2x)^{18} + \text{coeff. of } x^2 \text{ in } a(1-2x)^{18} + \text{coeff. of } x \text{ in } b(1-2x)^{18}$
 $\Rightarrow -^{18}C_3 \cdot 2^3 + a \cdot ^{18}C_2 \cdot 2^2 - b \cdot ^{18}C_1 \cdot 2 = 0$
 $17a - b = \frac{34 \times 16}{3} \dots (1)$
 similarly coeff. of $x^4 = ^{18}C_4 \cdot 2^4 - a \cdot ^{18}C_3 \cdot 2^3 + b \cdot ^{18}C_2 \cdot 2^2 = 0$
 $32a - 3b = 240 \dots (2)$
 solve (1) & (2) $\Rightarrow a = 16 ; b = \frac{272}{3}$

W.E-6: The coefficient of x^n in the expansion of $(1+x)(1-x)^n$ is

Sol. $(1+x)(1-x)^n = (1+x) \left({}^nC_0 - {}^nC_1x + {}^nC_2x^2 - \dots + (-1)^n \cdot {}^nC_nx^n \right)$
 coefficient x^n is $(-1)^n \cdot {}^nC_n + (-1)^{n-1} \cdot {}^nC_{n-1}$
 $= (-1)^n (1-n)$

W.E-7: The coefficient of x^7 in the expansion of $(1-x-x^2+x^3)^6$ is [AIEEE-2011]

Sol. $= (1-x)^6 (1-x^2)^6$
 coefficient of x^7
 $= (-^6C_1 \times -^6C_3) + (-^6C_3 \times +^6C_2) + (-^6C_5 \times -^6C_1)$
 $= 120 - 300 + 36 = -144$

➤ $(x+a)^n + (x-a)^n = 2 \left\{ {}^nC_0 x^n + {}^nC_2 x^{n-2} a^2 + {}^nC_4 x^{n-4} a^4 + \dots \right\}$
 ➤ $(x+a)^n - (x-a)^n = 2 \left\{ {}^nC_1 x^{n-1} a + {}^nC_3 x^{n-3} a^3 + {}^nC_5 x^{n-5} a^5 + \dots \right\}$

➤ (i) The number of terms in the expansion of $(x+a)^n + (x-a)^n$ is $\frac{(n+2)}{2}$ if n is even, $\frac{(n+1)}{2}$ if n is odd.

(ii) The number of terms in the expansion of $(x+a)^n - (x-a)^n$ is $n/2$ if n is even, $\frac{(n+1)}{2}$ if n is odd.

W.E-8: The number of terms in the expansion of $(x+2y)^{20} + (x-2y)^{20}$ is

Sol. The number of terms $= \frac{n}{2} + 1$ (where $n=20$)
 $= \frac{20}{2} + 1 = 11$

MULTINOMIAL THEOREM

➤ The expansion of $(a_1 + a_2 + a_3 + \dots + a_r)^n$ is called multinomial theorem

➤ The number of terms in

$$(a_1 + a_2 + a_3 + \dots + a_r)^n$$

is ${}^{n+r-1}C_{r-1} \quad (n \in N)$

W.E-9: The number of terms in the expansion of $(a+b+c+d)^n$ is

Sol. Put $r=4$ in above formulae $({}^{n+r-1}C_{r-1} \quad (n \in N))$
 No. of terms $= \frac{(n+1)(n+2)(n+3)}{3!}$

W.E-10: The number of distinct terms in $(a+b+c+d+e)^3$ is

Sol. $r=5, n=3$
 Number of terms $= {}^{n+r-1}C_{r-1} = 35$

➤ The general term in the expansion of

$$(x_1 + x_2 + \dots + x_p)^n \text{ is } \frac{n!}{n_1! n_2! \dots n_p!} x_1^{n_1} x_2^{n_2} \dots x_p^{n_p}$$

where $n_1 + n_2 + \dots + n_p = n$

➤ Coefficient of

$$x^p y^q z^r \text{ in } (ax + by + cz)^n \text{ is } \frac{n! a^p b^q c^r}{p! q! r!}$$

where $p + q + r = n$ & $n, p, q, r \in \mathbb{N}$

W.E-11: The coefficient of $a^3 b^4 c^5$ in the expansion of $(a + b + c)^{12}$ is

Sol. The coefficient of $a^3 b^4 c^5$ in the expansion of $(a + b + c)^{12} = \frac{12!}{3! 4! 5!} = 27720$

Middle term: If n is odd there will be two middle terms in the expansion of $(x + a)^n$, which

are $\left(\frac{n+1}{2}\right)^{th}$ and $\left(\frac{n+3}{2}\right)^{th}$ terms

➤ If n is even there will be only one middle term in the expansion of $(x + a)^n$, which is $\left(\frac{n}{2} + 1\right)^{th}$ term

➤ If n is odd. there are two greatest binomial coefficients in the expansion of $(x + a)^n$, which are ${}^n C_{\frac{n-1}{2}}$ and ${}^n C_{\frac{n+1}{2}}$ also ${}^n C_{\frac{n-1}{2}} = {}^n C_{\frac{n+1}{2}}$.

➤ If n is even, there is only one greatest binomial coefficient in the expansion of $(x + a)^n$, which is ${}^n C_{\frac{n}{2}}$

W.E-12: The middle term of $(1 - 3x + 3x^2 - x^3)^{2n}$ is

Sol. $((1 - x)^3)^{2n} = (1 - x)^{6n}$

$$\text{middle term} = \left(\frac{6n}{2} + 1\right)^{th} \text{ term} = {}^{6n} C_{3n} (-x)^{3n}$$

W.E-13: The coefficient of the middle term in the binomial expansion in powers of x of $(1 + \alpha x)^4$ and of $(1 - \alpha x)^6$ is the same if $\alpha =$

Sol. ${}^4 C_2 \alpha^2 = {}^6 C_3 (-\alpha)^3$; $\alpha = -\frac{3}{10}$

W.E-14: The middle term in the expansion of

$$\left(x - \frac{1}{x}\right)^{2n} \text{ is}$$

Sol. middle term = $\left(\frac{(2n)}{2} + 1\right)^{th}$ term

$$T_{n+1} = {}^{2n} C_n (-1)^n$$

$$\text{middle term} = \frac{1.3.5 \dots (2n-1)}{n!} (-2)^n$$

➤ If $n > 2, n \in \mathbb{N}$, then

$$(2n-1)^n + (2n)^n < (2n+1)^n$$

➤ If the coefficients of x^{r-1}, x^r, x^{r+1} in

$$(1+x)^n \text{ are in A.P. then } (n-2r)^2 = n+2$$

➤ If the coefficients of $r^{th}, (r+1)^{th}, (r+2)^{th}$ terms in the expansion of $(1+x)^n$ are in H.P, then $(n-2r)^2 + n = 0$

➤ Coefficient of x^{n-1} in

$$(x - \alpha_1)(x - \alpha_2) \dots (x - \alpha_n)$$

$$\text{is } -(\alpha_1 + \alpha_2 + \dots + \alpha_n)$$

➤ Coefficient of x^{n-1} in

$$(x + \alpha_1)(x + \alpha_2) \dots (x + \alpha_n)$$

$$\text{is } (\alpha_1 + \alpha_2 + \dots + \alpha_n)$$

➤ Coefficient of x^{n-2} in

$$(x - \alpha_1)(x - \alpha_2) \dots (x - \alpha_n)$$

$$\text{is } \frac{(\alpha_1 + \alpha_2 + \dots + \alpha_n)^2 - (\alpha_1^2 + \alpha_2^2 + \dots + \alpha_n^2)}{2}$$

Independent term: The term Independent

of x in the expansion of $\left(ax^p + \frac{b}{x^q}\right)^n$ is

$$T_{r+1} = {}^n C_r a^{n-r} b^r, \text{ where } r = \frac{np}{p+q} \text{ (integer)}$$

W.E-15: The term independent of x in the

expansion of $\left(\frac{x+1}{x^{2/3}-x^{1/3}+1}-\frac{x-1}{x-x^{1/2}}\right)^{10}$ is

Sol.
$$\left[\left(x^{1/3}+1\right)-\frac{x^{1/2}+1}{x^{1/2}}\right]^{10}$$
 [MAINS-2013]

$$=\left(x^{1/3}+1-1-x^{-1/2}\right)^{10}=\left(x^{1/3}-x^{-1/2}\right)^{10}$$

If $(r+1)^{\text{th}}$ term is the term independent of x

$$\text{then } r=\frac{10(1/3)}{1/3+1/2}=4$$

$\therefore$ The term independent of x is

$$T_5={}^{10}C_4(-1)^4={}^{10}C_4=210$$

➤ The term containing the coefficient of x^m in the

expansion of $\left(ax^p+\frac{b}{x^q}\right)^n$ is T_{r+1}

i.e, $T_{r+1}={}^nC_r a^{n-r} b^r$, where $r=\frac{np-m}{p+q}$ (integer)

➤ 'r'th term from end in the expansion of

$(x+y)^n$ is $(n-r+2)$ th term from beginning.

Numerically greatest term: Numerically greatest term in the expansion of $(1+x)^n$ where n is a positive integer.

a) If $\frac{(n+1)|x|}{1+|x|}=p$ (an integer), then p^{th} term and $(p+1)^{\text{th}}$ term are the two numerically greatest terms in $(1+x)^n$ Also $|t_p|=|t_{p+1}|$.

b) If $\frac{(n+1)|x|}{1+|x|}=p+F$ where p is an integer and $0 < F < 1$, then $(p+1)^{\text{th}}$ term is the numerically greatest term in $(1+x)^n$.

W.E-16: Numerically the greatest term in the

expansion of $(2+3x)^9$ when $x=\frac{3}{2}$, is

Sol. $(2+3x)^9=2^9\left(1+\frac{3x}{2}\right)^9$, $r=6$

$$\text{So, greatest term} = 2^9.T_{6+1}=\frac{7 \times 3^{13}}{2}$$

Binomial coefficients:

${}^nC_0, {}^nC_1, {}^nC_2, \dots, {}^nC_n$ are called the binomial coefficients in the expansion of $(x+a)^n$. They are denoted by $c_0, c_1, c_2, \dots, c_n$ respectively.

$c_0, c_1, c_2, \dots, c_n$ are the binomial coefficients in the expansion of $(1+x)^n$.

a) ${}^nC_0+{}^nC_1+{}^nC_2+{}^nC_3+\dots+{}^nC_n=2^n$

b) ${}^nC_0-{}^nC_1+{}^nC_2-{}^nC_3+\dots+(-1)^n.{}^nC_n=0$

c) ${}^nC_0+{}^nC_2+{}^nC_4+\dots+{}^nC_1+{}^nC_3+\dots=2^{n-1}$

d) ${}^nC_1+2.{}^nC_2+3.{}^nC_3+\dots+n.{}^nC_n=n.2^{n-1}$

e) $c_0+\frac{c_1}{2}x+\frac{c_2}{3}x^2+\dots+\frac{c_n}{n+1}x^n=\frac{(1+x)^{n+1}-1}{(n+1)x}$

f) $c_0+\frac{c_1}{2}+\frac{c_2}{3}+\frac{c_3}{4}+\dots+\frac{c_n}{n+1}=\frac{2^{n+1}-1}{n+1}$

g) $c_0-\frac{c_1}{2}+\frac{c_2}{3}-\frac{c_3}{4}+\dots+(-1)^n\frac{c_n}{n+1}=\frac{1}{n+1}$

h) $c_0+\frac{c_2}{3}+\frac{c_4}{5}+\dots=\frac{2^n}{n+1}$

i) $\frac{C_1}{2}+\frac{C_3}{4}+\frac{C_5}{6}+\dots=\frac{2^n-1}{n+1}$

j) a. $C_0+(a+d).C_1+(a+2d).C_2+\dots+(a+nd).C_n=(2a+nd)2^{n-1}$

k) a. $C_0-(a+d).C_1+(a+2d).C_2-\dots+(-1)^n(a+nd).C_n=0$

l) $C_0^2+C_1^2+C_2^2+\dots+C_n^2=2n.{}^nC_n$

m) $c_0^2-c_1^2+c_2^2-c_3^2+\dots+(-1)^n.c_n^2$

$$=\begin{cases} (-1)^{n/2}.{}^nC_{\frac{n}{2}}, & \text{if } n \text{ is even} \\ 0, & \text{if } n \text{ is odd} \end{cases}$$

n) a $C_0^2+(a+d).C_1^2+(a+2d).C_2^2+\dots+(a+nd).C_n^2=\left(\frac{2a+nd}{2}\right).2^n.C_n$

o) $c_0c_r+c_1c_{r+1}+\dots$

$$+c_{n-r}c_n=2^n.c_{n-r}=\frac{(2n)!}{(n-r)!(n+r)!}$$

p) ${}^m C_0 \cdot {}^n C_r + {}^m C_1 \cdot {}^n C_{r-1} + {}^m C_2 \cdot {}^n C_{r-2} + \dots + {}^m C_r \cdot {}^n C_0 = {}^{m+n} C_r$

q) $(C_0 + C_1)(C_1 + C_2) \dots (C_{n-1} + C_n) = \frac{(C_0 C_1 C_2 \dots C_{n-1})(n+1)^n}{n!}$

r) In the expansion of $(1+x)^n$,

$$\frac{C_1}{C_0} + 2 \frac{C_2}{C_1} + 3 \frac{C_3}{C_2} + \dots + n \frac{C_n}{C_{n-1}} = \frac{n(n+1)}{2}$$

- s) Let $f(x)$ be any polynomial in x
 a. Sum of the coefficients = $f(1)$
 b. Sum of the coefficients of even powers of

$$x = \frac{f(1) + f(-1)}{2}$$

- c. Sum of the coefficients of odd powers of

$$x = \frac{f(1) - f(-1)}{2}$$

W.E-17: If $(1+x-2x^2)^6 = 1 + a_1x + a_2x^2 + \dots + a_{12}x^{12}$

then $a_2 + a_4 + a_6 + \dots + a_{12}$ has the value

Sol. put $x = 1$ and $x = -1$ and adding these results.

$$a_2 + a_4 + a_6 + \dots + a_{12} = 31$$

W.E-18: The sum of the series

$${}^{20}C_0 - {}^{20}C_1 + {}^{20}C_2 - {}^{20}C_3 + \dots - \dots + {}^{20}C_{10} \text{ is}$$

[AIEEE-2007]

Sol. $(1+x)^{20} = {}^{20}C_0 + {}^{20}C_1x + {}^{20}C_2x^2 + {}^{20}C_3x^3 + \dots + {}^{20}C_{20}x^{20}$

put $x = -1$

$$\Rightarrow 0 = 2 \left[{}^{20}C_0 - {}^{20}C_1 + {}^{20}C_2 - {}^{20}C_3 + \dots - \dots + {}^{20}C_{10} \right] - {}^{20}C_{10}$$

$$\therefore {}^{20}C_0 - {}^{20}C_1 + {}^{20}C_2 - {}^{20}C_3 + \dots - \dots + {}^{20}C_{10}$$

$$= \frac{1}{2} {}^{20}C_{10}$$

W.E-19: If the sum of the coefficients in the expansion of $(1-3x+10x^2)^n$ is a and if the sum of the coefficients in the expansion of $(1+x^2)^n$ is b , then

Sol. Put $x=1$, in both expansions.

$$\text{i.e., } 8^n = a \text{ and } 2^n = b \quad \therefore a = b^3$$

W.E-20: If a_k is the coefficient of x^k in the expansion of $(1+x+x^2)^n$ for $k = 0, 1, 2, \dots, 2n$ then $a_1 + 2a_2 + 3a_3 + \dots + 2na_{2n} =$ [EAM-2007]

Sol. Differentiating w.r.t x , put $x = 1$

$$a_1 + 2a_2 + 3a_3 + \dots + 2na_{2n} = n \cdot 3^n$$

- The coefficient of x^r in $(1+x)^m (1+x)^n$ is $(m+n)C_r$
- The coefficient of the term independent of x in the expansion of $(1+x)^m \left(1 + \frac{1}{x}\right)^n$ is $m + nC_n$

Greatest Term: If the greatest term in the expansion of $(1+x)^{2n}$ has also greatest coefficient,

$$\text{then } \frac{n}{n+1} < x < \frac{n+1}{n}$$

the greatest term in the expansion of $(1+x)^{2n-1}$

has also greatest coefficient, then $\frac{n-1}{n+1} < x < \frac{n+1}{n-1}$

BINOMIAL THEOREM FOR RATIONAL INDEX

- The no. of rational terms in the expansion of $\left(a^{\frac{1}{l}} + b^{\frac{1}{k}}\right)^n$ is $\left\lceil \frac{n}{L.C.m \text{ of } \{l, k\}} \right\rceil$ greatest integer function when none of l and k is a factor of 'n' and when at least one of l and k is a factor of n is $\left\lceil \frac{n}{L.C.m \text{ of } \{l, k\}} \right\rceil + 1$ where $[.]$ greatest integer function.

W.E-21: The number of irrational terms in

$$\left(\sqrt[8]{5} + \sqrt[6]{2}\right)^{100} \text{ is}$$

Sol. $T_{r+1} = {}^{100}C_r \left(\sqrt[8]{5}\right)^{100-r} \left(\sqrt[6]{2}\right)^r = {}^{100}C_r 5^{(100-r)/8} \cdot 2^{r/6}$

$$\therefore r = 12, 36, 60, 84$$

The number of rational terms = 4

No. of irrational terms = $101 - 4 = 97$

W.E-22: If n is a positive integer, then

$$\left(\sqrt{3} + 1\right)^{2n} - \left(\sqrt{3} - 1\right)^{2n} \text{ is [AIEEE-2012]}$$

$$\left(\sqrt{3} + 1\right)^{2n} - \left(\sqrt{3} - 1\right)^{2n}$$

Sol. $= 2 \left[{}^{2n}C_1 (\sqrt{3})^{2n-1} + {}^{2n}C_3 (\sqrt{3})^{2n-3} + \dots \right]$

is an irrational number

Binomial Theorem for Negative Index

➤ If n is a rational number and $-1 < x < 1$, then

a. $(1+x)^n = 1 + nx + \frac{n(n-1)}{2}x^2 + \frac{n(n-1)(n-2)}{3}x^3 + \dots + \frac{n(n-1)(n-2)\dots(n-r+1)}{r!}x^r + \dots\infty$

b. $(1-x)^n = 1 - nx + \frac{n(n-1)}{2}x^2 - \dots + (-1)^r \frac{n(n-1)(n-2)\dots(n-r+1)}{r!}x^r + \dots\infty$

c. $(1-x)^{-n} = 1 + nx + \frac{n(n+1)}{2}x^2 + \frac{n(n+1)(n+2)}{3}x^3 + \dots + \frac{n(n+1)(n+2)\dots(n+r-1)}{r!}x^r + \dots\infty$

d. $(1+x)^{-n} = 1 - nx + \frac{n(n+1)}{2!}x^2 - \frac{n(n+1)(n+2)}{3!}x^3 + \dots + (-1)^r \frac{n(n+1)(n+2)\dots(n+r-1)}{r!}x^r + \dots\infty$

➤ $(1-x)^{-1} = 1 + x + x^2 + x^3 + \dots + x^r + \dots\infty$

➤ $(1+x)^{-1} = 1 - x + x^2 - x^3 + \dots + (-1)^r x^r + \dots\infty$

➤ $(1-x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + \dots + (r+1)x^r + \dots\infty$

➤ $(1+x)^{-2} = 1 - 2x + 3x^2 - 4x^3 + \dots + (-1)^r (r+1)x^r + \dots\infty$

➤ $(1-x)^{-3} = 1 + 3x + 6x^2 + 10x^3 + \dots$

$$+ \frac{(r+1)(r+2)}{r!}x^r + \dots\infty$$

➤ $(1+x)^{-3} = 1 - 3x + 6x^2 - 10x^3 + \dots$

$$+ (-1)^r \frac{(r+1)(r+2)}{r!}x^r + \dots\infty$$

➤ Number of terms in $(1+x)^n$ is

(i) 'n+1' when 'n' positive integer.

(ii) infinite when 'n' is not a positive integer & $|x| < 1$

➤ First negative term in $(1+x)^{p/q}$ when $0 < x < 1$, p, q are positive integers & 'p' is not a multiple of 'q' is $T_{\left[\frac{p}{q}\right]+3}$

W.E-23: If x is positive, the first negative term in the expansion of $(1+x)^{27/5}$ is

Sol. Required Term

$$= T_{\left[\frac{27}{5}\right]+3} = T_8$$

T_8 is the first negative term

➤ **Approximations:** If 'x' is very small so that x^2 and higher powers of 'x' can be neglected. Then

$$(1+x)^n \cong 1 + nx$$

W.E-24: If $(a+bx)^{-3} = \frac{1}{27} + \frac{1}{3}x + \dots$ then the ordered pair (a, b) = [EAMCET-2014]

Sol. $a^{-3} \left(1 + \frac{b}{a}x\right)^{-3} = \frac{1}{27} + \frac{1}{3}x + \dots$

$$\Rightarrow \frac{1}{a^3} \left[1 - \frac{3b}{a}x + \dots\right] = \frac{1}{27} + \frac{1}{3}x + \dots$$

$$\Rightarrow a^3 = 27, \frac{-3b}{a^4} = \frac{1}{3} ; (a, b) = (3, -9)$$

W.E-25: If x is so small that x^3 and higher powers

of x may be neglected, then $\frac{(1+x)^{3/2} - \left(1 + \frac{1}{2}x\right)^3}{(1-x)^{1/2}}$

may be approximated as

Sol.
$$= \frac{\left(1 + \frac{3}{2}x + \frac{3/4}{2!}x^2\right) - \left(1 + \frac{3}{2}x + \frac{3}{4}x^2\right)}{(1-x)^{1/2}}$$

$$= \frac{-\frac{3}{8}x^2(1-x)^{-1/2}}{1-x} = -\frac{3}{8}x^2$$

- $1 + x + x^2 + \dots + x^n = \frac{1-x^{n+1}}{1-x}$ where $x \neq 1$
- If $n_{c_{r-1}} : n_{c_r} : n_{c_{r+1}} = a : b : c$ then

$$r = \frac{a(b+c)}{b^2-ac}, \quad n = \frac{ab+bc+2ca}{b^2-ac}$$
- For $n > 6$, $\left(\frac{n}{3}\right)^n < n! < \left(\frac{n}{2}\right)^n$
- $n^{n+1} > (n+1)^n, \forall n \in \mathbb{N}$
- $n^n > (n+1)^{n-1}, \forall n \in \mathbb{N}$
- **Divisibility:** $(1+\alpha)^n - 1$ is divisible by α

W.E-26: The remainder left out when

$8^{2n} - (62)^{2n+1}$ is divided by 9 is [AIEEE-2009]

Sol. $8^{2n} - (62)^{2n+1} = (9-1)^{2n} - (63-1)^{2n+1}$

$$= \left[{}^{2n}C_0 9^{2n} + {}^{2n}C_1 9^{2n-1}(-1) + \dots + {}^{2n}C_{2n-1} 9(-1)^{2n-1} + 1 \right]$$

$$- \left[{}^{2n+1}C_0 (63)^{2n+1} - {}^{2n+1}C_1 (63)^{2n} + \dots - {}^{(2n+1)}C_{2n} (63) - 1 \right]$$

so remainder is $1 - (-1) = 2$

- $(1+\alpha)^n - n\alpha - 1$ is divisible by α^2
- $(1+\alpha)^n - {}^nC_2 \alpha^2 - n\alpha - 1$ is divisible by α^3
- $1^3.C_1 - 2^3.C_2 + 3^3.C_3 - 4^3.C_4 + \dots + (-1)^n n^3.C_n = 0$

➤ $1^3.C_1 - 2^3.C_2 + 3^3.C_3 - 4^3.C_4 + \dots + (-1)^n n^3.C_n = 0$

➤ $\sum_{r=0}^n r(r-1)n_{C_r} = n(n-1)2^{n-2}$

➤ $\sum_{r=0}^n r(r-1)(r-2)n_{C_r} = n(n-1)(n-2)2^{n-3}$

➤ If $(1+x)^n = C_0 + C_1x + C_2x^2 + \dots + C_nx^n$

Then $\sum_{0 \leq i < j \leq n} (i+j)C_iC_j = n(2^{2n-1} - \frac{1}{2}2n_{C_n})$



C.U.Q

1. If n is a positive integer, then the number of terms in the expansion of $(x+a)^n$ is
 1) n 2) $n+1$ 3) $n+2$ 4) infinitely many
2. The number of terms in the expansion of $(x+y+z)^n$ is
 1) $\frac{n(n+1)}{2}$ 2) $\frac{(n+1)(n+2)}{2!}$
 3) $\frac{n(n+3)}{2}$ 4) $\frac{(n+1)(n+3)}{2}$
3. If the coefficient of x^n in $(1+x)^{2n}$ is 'a' and the coefficient of x^n in $(1+x)^{2n-1}$ is b, then $\frac{a}{b} =$
 1) $2n+1$ 2) $4n+1$ 3) $2n$ 4) n
4. If the middle term of $(1+x)^{2n}$ is the greatest term, then x lies between
 1) $n-1 < x < n$ 2) $\frac{n}{n+1} < x < \frac{n+1}{n}$
 3) $n < x < n+1$ 4) $\frac{n+1}{n} < x < \frac{n}{n+1}$
5. If n is a positive integer, $\sum_{r=0}^n ({}^nC_r)^2 =$
 1) 0 2) ${}^nC_{n/2}$ 3) $\frac{(2n)!}{n!}$ 4) $\frac{(2n)!}{(n!)^2}$
6. In $(1+ax)^n$, sum of the coefficients is S_1 . If we double 'a' and half 'n', the new sum is S_2 . Then
 1) $S_1 > S_2$ 2) $S_1 < S_2$
 3) $S_1 = S_2$ 4) Cannot be decided

7. In the binomial expansion of $(a-b)^n, n \geq 5$, the sum of 5th and 6th terms is zero then a/b equal to [AIEEE - 2008]

1) $\frac{5}{n-4}$ 2) $\frac{6}{n-5}$ 3) $\frac{n-5}{6}$ 4) $\frac{n-4}{5}$

8. $\binom{2n}{0}^2 - \binom{2n}{1}^2 + \binom{2n}{2}^2 - \dots + \binom{2n}{2n}^2 =$

1) $2^n C_n$ 2) $(-1)^n \cdot 2^n C_n$ 3) $(-1)^{n/2} \cdot n C_{n/2}$ 4) $(-1)^{n/2} \cdot 2^n C_n$

9. The coefficient of x^p in the expansion of

$\left(x^2 + \frac{1}{x}\right)^{2n}$, when exists is

1) $2^n C_{\frac{4n+p}{3}}$ 2) $2^n C_{\frac{2n+p}{3}}$

3) $2^n C_{\frac{n+p}{3}}$ 4) $n C_{\frac{n+p}{3}}$

10. If a term independent of x exist in the

expansion of $\left(x + \frac{1}{x^2}\right)^n$, then n must be

1) a multiple of 2 2) a multiple of 3
3) a multiple of 5 4) a multiple of 7

11. In the expansion of $(a+b)^n$ if two consecutive

terms are equal, then $\frac{(n+1)b}{a+b}$ and $\frac{(n+1)a}{a+b}$ are

1) integers 2) complex numbers
3) irrational numbers 4) does not exist

12. In the expansion of $(1+x)^n$ if the coefficients of three consecutive terms are in A.P. then $n+2$ is

1) n^2 2) a perfect square
3) a perfect cube 4) n^3

13. If the coefficient of x^{2r} in the expansion of

$\left(x + \frac{1}{x^2}\right)^{n-3}$ is not zero, then $\left(\frac{n-2r}{3}\right)$ is

1) a rational number 2) a positive integer
3) a negative integer 4) a positive rational number

14. $m C_r + m C_{r-1} \cdot n C_1 + m C_{r-2} \cdot n C_2 + \dots + n C_r =$

1) $m + n C_{n+r}$ 2) $m + n C_{m+r}$ 3) $(m+n) C_r$ 4) 0

15. The coefficient of $\frac{1}{x}$ in the expansion of

$(1+x)^n \left(1 + \frac{1}{x}\right)^n$ is

1) $\frac{n!}{(n-1)!(n+1)!}$ 2) $\frac{2n!}{(n-1)!(n+1)!}$

3) $\frac{n!}{(2n-1)!(2n+1)!}$ 4) $\frac{2n!}{(2n-1)!(2n+1)!}$

16. The ratio of $(r+1)$ th and $(r-1)$ th terms in the expansion of $(a-b)^n$ is

1) $\frac{(n-r+2)(n-r+1)}{r(r-1)} \cdot \frac{b^2}{a^2}$

2) $\frac{(n-r+2)(n-r+1)}{r(r-1)} \cdot \frac{a^2}{b^2}$

3) $\left(\frac{n-r+2}{r}\right) \cdot \frac{b}{a}$ 4) $\left(\frac{n-r+1}{r-1}\right) \cdot \frac{b}{a}$

17. If a, b, c, d are any four consecutive coefficients in the expansion of $(1+x)^n$, then

$\frac{a}{a+b}, \frac{b}{b+c}, \frac{c}{c+d}$ are in

1) A.P. 2) G.P. 3) H.P. 4) A.G.P

18. If $(1+x)^n = \sum_{i=0}^n C_i x^i$, then the sum of the products of C_i 's taken two at a time is

represented by $\sum_{0 \leq i < j \leq n} C_i C_j$

1) $2^n - \frac{(2n)!}{2(n!)^2}$ 2) $2^n - \frac{(2n)!}{2(n!)^2}$

3) $\frac{1}{2} \left(2^{2n} - \frac{(2n)!}{(n!)^2} \right)$ 4) $\frac{2^{2n}}{2(n!)^2}$

19. If $(1+x)(1+x+x^2)(1+x+x^2+x^3) \dots$

$(1+x+x^2+\dots+x^n) = a_0 + a_1 x + a_2 x^2 + \dots + a_m x^m$

Then the value of

$a_0 - a_1 + a_2 - a_3 + \dots + (-1)^m a_m$ is

1) $(n-2)$ 2) $(m-1)$ 3) 0 4) mn

20. The sum of $r_{C_r} + (r+1)_{C_r} + (r+2)_{C_r} + \dots + n_{C_r}$ ($n \geq r$)
 1) $n_{C_{r+1}}$ 2) $n+1_{C_{r+1}}$ 3) $n+1_{C_r}$ 4) $n+1_{C_{r-1}}$
21. The digit in the units place of the number $183! + 3^{183}$ is
 1) 0 2) 3 3) 6 4) 7
22. The last digit in 7^{100} is
 1) 1 2) 3 3) 7 4) 9
23. The coefficient of $a^4 b^3 c^2 d$ in the expansion of $(a+b+c+d)^{10}$ is
 1) 12600 2) 16200 3) 21600 4) 26100
24. If T_r denotes the r^{th} term in the expansion of $\left(x + \frac{1}{x}\right)^{23}$, then
 1) $T_{12} = T_{13}$ 2) $T_{13} = x^2 T_{12}$
 3) $T_{12} = x^2 T_{13}$ 4) $T_{12} + T_{13} = 25$
25. In the expansion of $\left(x + x^{\log_{10} x}\right)^5$ the third term is 10^6 then x
 1) 1 2) 2 3) 10 4) 100
26. The value of $(1.03)^{\frac{1}{3}}$ up to 4 decimal places is
 1) 1.0998 2) 1.0099 3) 1.0098 4) 1.0989
27. The binomial coefficients which are in decreasing order
 1) $15_{C_5}, 15_{C_6}, 15_{C_7}$ 2) $15_{C_{10}}, 15_{C_8}, 15_{C_9}$
 3) $15_{C_6}, 15_{C_7}, 15_{C_8}$ 4) $15_{C_7}, 15_{C_6}, 15_{C_5}$

C.U.Q - KEY

- 1) 2 2) 2 3) 1 4) 2 5) 4 6) 1 7) 4
 8) 2 9) 2 10) 2 11) 1 12) 2 13) 2 14) 3
 15) 2 16) 1 17) 1 18) 3 19) 3 20) 2 21) 4
 22) 1 23) 1 24) 3 25) 3 26) 2 27) 4

C.U.Q - HINTS

1. n is positive integer, number of terms = $n+1$
 2. $n+r-1_{C_{r-1}}$
 3. $\frac{{}^{2n}C_n}{{}^{2n-1}C_n} = 2$
 4. Middle term T_{n+1} ; $n < \frac{(2n+1)|x|}{|x|+1} < n+1$

5. $C_0^2 + C_1^2 + C_2^2 + \dots + C_n^2 = {}^{2n}C_n$
 6. Substitute n, a , values and verify
 7. Given $T_5 + T_6 = 0 \Rightarrow {}^nC_4 a^{n-4} b^4 - {}^nC_5 a^{n-5} b^5 = 0$
 8. substitute a positive integer for n and verify
 9. $r = \frac{4n-p}{3}$
 Coefficient of x^p in $\left(x^2 + \frac{1}{x}\right)^{2n}$ is ${}^{2n}C_{\frac{4n-p}{3}}$
 10. Use the formula $r = \frac{np}{p+q}$
 11. $\frac{(n+1)\left|\frac{b}{a}\right|}{\left|\frac{b}{a}\right|+1}$ and $\frac{(n+1)\left|\frac{a}{b}\right|}{\left|\frac{a}{b}\right|+1}$ are integers
 12. $(n-2r)^2 = n+2$
 13. $\frac{(n-3)1-2r}{3} \neq 0$
 $\frac{n-2r}{3} - 1 \neq 0 \Rightarrow \frac{n-2r}{3}$ Should be an integer
 14. Vandermonde's theorem $(m+n)C_r$
 15. Coefficient of $\frac{1}{x}$ in $(1+x)^n \left(1 + \frac{1}{x}\right)^n =$
 Coefficient of x^{n-1} in $(1+x)^{2n}$
 16. $\frac{T_{r+1}}{T_{r-1}}$
 17. ${}^nC_r = a, {}^nC_{r+1} = b, {}^nC_{r+2} = c, {}^nC_{r+3} = d$
 substitute above in the expansion and simplify
 18. Standard result
 19. Put $x=-1$
 20. $n_{C_{r-1}} + n_{C_r} = n+1_{C_r}$
 21. $3^1 = 3, 3^2 = 9, 3^3 = 27, 3^4 = 81 \dots 3^{183} = (3^4)^{45} \cdot 3^3$
 and in $183!$ unit place is '0'
 22. $7^1 = 7, 7^2 = 49, 7^3 = 343, 7^4 = 2401$ and $7^{100} = (7^4)^{25}$
 23. $(-1)^4 \frac{10!}{4!3!2!1!}$

24. $T_{12} = 23_{C_{11}} x^{12} \frac{1}{x^{11}}, T_{13} = 23_{C_{12}} x^{11} \frac{1}{x^{12}}$
25. $T_3 = 10^6 \Rightarrow 5_{C_2} x^3 \left(x^{\log_{10} 5} \right)^2 = 10^6, x=10$
26. $(1+0.03)^{\frac{1}{3}}$ expand
27. $n_{C_r} = \frac{n!}{(n-r)!r!}$



LEVEL-I (C.W)

NO.OF TERMS & PARTICULAR TERM

1. The number of terms in the expansion of $(1+x)^{21}$ is
1) 20 2) 21 3) 22 4) 24
2. The 4th term in the expansion of $\left(\sqrt{x} + \frac{1}{x}\right)^{12}$ is
1) $110x^{\frac{3}{2}}$ 2) $220x^{\frac{3}{2}}$ 3) $220x^2$ 4) $110x^2$
3. The $(n+1)^{\text{th}}$ term from the end in $\left(x - \frac{1}{x}\right)^{3n}$ is
1) ${}^{3n}C_n \cdot x^{-n}$ 2) $(-1)^n \cdot {}^{3n}C_n \cdot x^{-n}$
3) ${}^{3n}C_n \cdot x^n$ 4) $(-1)^n \cdot {}^{3n}C_n \cdot x^n$

PROBLEMS BASED ON COEFFICIENT OF x^k

4. If the coefficient of x in $\left(x^2 + \frac{k}{x}\right)^5$ is 270, then $k=$
1) 3 2) 4 3) 5 4) 6
5. The coefficients of x^p and x^q (p and q are positive integers) in the expansion of $(1+x)^{p+q}$ are
1) equal
2) equal with opposite signs
3) reciprocal to each other 4) unequal
6. If the coefficients of 5th, 6th and 7th terms in the expansion of $(1+x)^n$ are in A.P. then $n =$
1) 7 2) 14 3) 7 or 14 4) 8
7. The coefficient of x^9 in $(1+9x+27x^2+27x^3)^6$ is
1) ${}^{18}C_9 \cdot 3^9$ 2) ${}^{18}C_8 \cdot 3^9$ 3) ${}^{18}C_{10} \cdot 3^8$ 4) ${}^8C_9 \cdot 3^7$
8. If the coefficient of x in the expansion of $(1+ax)^8(1+3x)^4 - (1+x)^3(1+2x)^4$ is zero, then $a=$
1) $1/4$ 2) $-1/4$ 3) $1/8$ 4) $-1/8$

INDEPENDENT TERM & MIDDLE TERM

9. The term independent of x in the expansion of $(1+x)^{10} \cdot \left(1 + \frac{1}{x}\right)^{10}$ is
1) ${}^{10}C_5$ 2) ${}^{20}C_{10}$ 3) ${}^{20}C_5$ 4) ${}^{10}C_2$
10. If the fourth term in the expansion of $\left(px + \frac{1}{x}\right)^n$ is $5/2$, then $(n, p) =$
1) $(3, 1/2)$ 2) $(6, 1/2)$ 3) $(5, 1/2)$ 4) $(6, 2)$
11. The term independent of x in the expansion of $\left(\sqrt{\frac{x}{3}} + \frac{\sqrt{3}}{x^2}\right)^{10}$ is
1) $5/9$ 2) $5/3$ 3) $1/3$ 4) $4/3$
12. The middle term of $\left(x - \frac{1}{x}\right)^{2n+1}$ is
1) ${}^{2n+1}C_n \cdot x$ 2) ${}^{2n+1}C_n$
3) $(-1)^n \cdot {}^{2n+1}C_n$ 4) $(-1)^n \cdot {}^{2n+1}C_n \cdot x$

PROBLEMS BASED ON RATIOS & NUMERICALLY GREATEST TERM

13. In the expansion of $\left(\sqrt{a} + \frac{1}{\sqrt{3a}}\right)^n$ if the ratio of the binomial coefficient of the 4th term to the binomial coefficient of the 3rd term is $\frac{10}{3}$, the 5th term is
1) $55a$ 2) $45a^2$ 3) $50a^2$ 4) $55a^2$
14. The two successive terms in the expansion of $(1+x)^{24}$ whose coefficients are in the ratio 4 : 1 are
1) t_{18}, t_{19} 2) t_{19}, t_{20} 3) t_{20}, t_{21} 4) t_{21}, t_{22}
15. When $x = \frac{5}{2}$, numerically greatest term in the expansion of $(3+2x)^{15}$ is
1) 6th 2) 8th 3) 10th 4) 12th

PROBLEMS BASED ON COEFFICIENTS

16. The coefficients of x^{16} and x^{17} in $(a+bx)^{101}$ are equal. If a and b are natural numbers, then the least possible value of 'a' is
1) 1 2) 3 3) 5 4) 7

17. The coefficient of x^k ($0 \leq k \leq n$) in the expansion of $1+(1+x)+(1+x)^2+\dots+(1+x)^n$ is
 1) $(n+1)C_k$ 2) nC_k 3) ${}^nC_{k+1}$ 4) $(n+1)C_{k+1}$
18. The 21st and 22nd terms in the expansion of $(1+x)^{44}$ are equal. Then $x=$
 1) $8/7$ 2) $7/8$ 3) 7 4) 8
19. Sum of the coefficients of $(1-x)^{25}$ is
 1) -1 2) 1 3) 0 4) 2^{25}
20. Coefficient of x^{10} in $(1+2x^4)(1-x)^8$ is
 1) -56 2) 56 3) 112 4) -112

NUMBER OF TERMS

21. The total number of terms in the expansion of $(x+a)^{100}+(x-a)^{100}$ is
 1) 202 2) 51 3) 101 4) 50
22. The number of rational terms in the expansion of $(\sqrt{3} + \sqrt[4]{5})^{124}$ is
 1) 31 2) 32 3) 33 4) 34

BINOMIAL COEFFICIENTS

23. If n is odd, ${}^nC_1 + {}^nC_3 + {}^nC_5 + \dots + {}^nC_{2[n/2]-1} =$
 where $[]$ denotes greatest integer
 1) 2^{n-1} 2) $2^{n-1}-1$ 3) 2^n 4) 2^{n-1}
24. If $n \geq 2$ then
 $3.C_1 - 4.C_2 + 5.C_3 - \dots + (-1)^{n-1}(n+2).C_n =$
 1) -1 2) 2 3) -2 4) 1
25. $2.C_0 + 2^2 \cdot \frac{C_1}{2} + 2^3 \cdot \frac{C_2}{3} + \dots + 2^{n+1} \cdot \frac{C_n}{n+1} =$
 1) $\frac{3^{n+1}-1}{2(n+1)}$ 2) $\frac{3^{n+1}-1}{n+1}$ 3) $\frac{3^n-1}{n+1}$ 4) $\frac{3^n+1}{n+1}$
26. $C_0^2 + 3.C_1^2 + 5.C_2^2 + \dots + (2n+1).C_n^2 =$
 1) $(n+1)2^n$ 2) $(2n+1)2^n C_n$
 3) $(n+1).2^n C_n$ 4) $(2n-1).2^n C_n$
27. $\frac{C_0}{1.2} + \frac{C_1}{2.3} + \frac{C_2}{3.4} + \dots + \frac{C_n}{(n+1)(n+2)} =$
 1) $\frac{2^{n+1}-n}{(n+1)(n+2)}$ 2) $\frac{2^{n+2}-n-3}{(n+1)(n+2)}$
 3) $\frac{2^n-1}{n+1}$ 4) $\frac{2^{n+1}+1}{(n+1)(n+2)}$

28. $C_1 + 2C_2.a + 3.C_3.a^2 + \dots + 2n.C_{2n}.a^{2n-1} =$
 1) $n(1+a)^{n-1}$ 2) $n(1+a)^n$
 3) $2n(1+a)^{2n-1}$ 4) $2n(1+a)^{2n}$
29. If $(x-2)^{100} = \sum_{r=0}^{100} a_r \cdot x^r$, then $a_1 + 2a_2 + \dots + 100a_{100} =$
 1) 100 2) -100 3) 1 4) 101
30. If $(1+x+x^2)^n = \sum_{r=0}^{2n} a_r x^r$, then
 $a_1 - 2a_2 + 3a_3 - \dots - 2n.a_{2n} = \dots$
 1) 0 2) 1 3) n 4) $-n$

PROBLEMS BASED ON SUM OF COEFFICIENTS

31. If the sum of the coefficients in the expansion of $(x+y)^n$ is 4096 , then the greatest coefficient is
 1) ${}^{11}C_5$ 2) ${}^{12}C_5$ 3) ${}^{12}C_6$ 4) ${}^{14}C_7$
32. In the expansion of $(x+a)^n$, sum of the odd terms is P and the sum of the even terms is Q , then $4PQ =$
 1) $(x^2-a^2)^n$ 2) $(x+a)^{2n} + (x-a)^{2n}$
 3) $(x+a)^{2n} - (x-a)^{2n}$ 4) $(x-a)^{2n} - (x+a)^{2n}$

BINOMIAL THEOREM FOR RATIONAL INDEX

33. Expansion of $(4-7x)^{-\frac{2}{5}}$ is valid if
 1) $x < \frac{4}{7}$ 2) $x > \frac{4}{7}$
 3) $-\frac{4}{7} < x < \frac{4}{7}$ 4) $-\frac{4}{7} \leq x \leq \frac{4}{7}$
34. In the expansion of $(2-3x)^{-1}$, the 3rd term is
 1) $\frac{9x^2}{4}$ 2) $\frac{9x^2}{8}$ 3) $\frac{27x^3}{8}$ 4) $\frac{27x^3}{16}$
35. The coefficient of x^n in $\frac{(1+x)^2}{(1-x)^3}$ is
 1) $3n^2+2n+1$ 2) $2n^2+2n+1$
 3) $3n^2+n+1$ 4) $2n^2-2n+1$
36. Coefficient of x^4 in the expansion of $\frac{1}{(x+1)(x+2)}$ is
 1) $\frac{1}{32}$ 2) $\frac{11}{32}$ 3) $\frac{21}{32}$ 4) $\frac{31}{32}$

37. Coefficient of x^n in the expansion of $(1-2x+3x^2-4x^3+\dots\infty)^2$ is

- 1) $\frac{n(n+1)(n+2)}{3!}$ 2) $(-1) \frac{n(n+1)(n+2)}{3!}$
 3) $\frac{(n+1)(n+2)(n+3)}{3!}$
 4) $(-1)^n \frac{(n+1)(n+2)(n+3)}{3!}$

38. If x is small, then $\sqrt{x^2+16} - \sqrt{x^2+9} =$

- 1) $1 + \frac{x^2}{24}$ 2) $1 - \frac{x^2}{24}$ 3) $1 + \frac{x^2}{48}$ 4) $1 - \frac{x^2}{48}$

39. $\frac{7}{5} \left(1 + \frac{1}{10^2} + \frac{1.3}{1.2} \cdot \frac{1}{10^4} + \frac{1.3.5}{1.2.3} \cdot \frac{1}{10^6} + \dots \right) =$

- 1) $\sqrt{2}$ 2) $2\sqrt{2}$ 3) $2^{1/3}$ 4) $\sqrt{\frac{2}{3}}$

40. If $z = \frac{1}{3} + \frac{1.3}{3.6} + \frac{1.3.5}{3.6.9} + \dots$ then

- 1) $z^2 - 2z + 2 = 0$ 2) $z^2 - 2z - 2 = 0$
 3) $z^2 + 2z - 2 = 0$ 4) $z^2 + 2z + 2 = 0$

41. If $|x| < 1$ then $1 + n \left(\frac{2x}{1+x} \right) + \frac{n(n+1)}{2!} \left(\frac{2x}{1+x} \right)^2 + \dots \infty$

- 1) $\left(\frac{1+x}{1-x} \right)^n$ 2) $\left(\frac{1-x}{1+x} \right)^n$ 3) $\left(\frac{1+x}{2x} \right)^n$ 4) $\left(\frac{2x}{1+x} \right)^n$

42. $1 - \frac{1}{5} + \frac{1.4}{5.10} - \frac{1.4.7}{5.10.15} + \dots \infty =$

- 1) $\sqrt[3]{5}$ 2) $\frac{1}{2} \sqrt[3]{5}$ 3) $\left(\frac{5}{2} \right)^{\frac{1}{3}}$ 4) $\frac{1}{2} \left(\frac{5}{2} \right)^{\frac{1}{3}}$

43. The sum of the series

$$1 + \frac{k}{3} + \frac{k(k+1)}{3.6} + \frac{k(k+1)(k+2)}{3.6.9} + \dots \text{ is}$$

- 1) $\left(\frac{2}{3} \right)^4$ 2) $\left(\frac{3}{2} \right)^k$ 3) $\frac{2}{3}$ 4) $\frac{3}{2}$

BINOMIAL THEOREM APPLICATIONS

44. which is larger $(1.01)^{1000000}$ or 10000

- 1) $(1.01)^{1000000}$ 2) 10000
 3) can't say 4) both are equal

45. Sum of the coefficients of all integral power of x is $(1+2\sqrt{x})^{40}$ is

- 1) $\frac{3^{40}-1}{2}$ 2) $\frac{3^{40}+1}{2}$ 3) $\frac{3^{38}-1}{2}$ 4) $\frac{3^{38}+1}{2}$

46. If c is small in comparison with l then

$$\left(\frac{l}{l+c} \right)^{\frac{1}{2}} + \left(\frac{l}{l-c} \right)^{\frac{1}{2}} =$$

- 1) $2 + \frac{3c}{4l}$ 2) $2 + \frac{3c^2}{4l^2}$ 3) $l + \frac{3c^2}{4l^2}$ 4) $l + \frac{3c}{4l}$

47. If the number of terms in the expansion of $(x+y+z)^n$ are 36, then the value of n

- 1) 7 2) 9 3) 4 4) 2

48. Coefficient of x^7 in the expansion of $(1+3x-2x^3)^{10}$.

- 1) 62640 2) 64620 3) 65640 4) 62330

LEVEL-I (C.W) - KEY

- 1) 3 2) 2 3) 1 4) 1 5) 1 6) 3 7) 1
 8) 4 9) 2 10) 2 11) 2 12) 4 13) 4 14) 3
 15) 3 16) 3 17) 4 18) 2 19) 3 20) 2 21) 2
 22) 2 23) 2 24) 2 25) 2 26) 3 27) 2 28) 3
 29) 2 30) 4 31) 3 32) 3 33) 3 34) 2 35) 2
 36) 4 37) 4 38) 2 39) 1 40) 3 41) 1 42) 2
 43) 2 44) 1 45) 2 46) 2 47) 1 48) 1

LEVEL-I (C.W) - HINTS

- $n+1$
- Use the formula $T_{r+1} = {}^nC_r x^{n-r} a^r$
- $(3n - (n+1) + 2)^{th}$ term from beginning
- $r = \frac{np-m}{p+q}$
- Coefficient of $x^p = {}^{p+q}C_p$
 coefficient of $x^q = {}^{p+q}C_q$
- $n^2 - (4r+1)n + 4r^2 - 2 = 0$ where $r = 5$
- Coefficient of x^9 in $(1+3x)^{18} = {}^{18}C_9 3^9$
- Coefficient of x in $(1+ax)^8 (1+3x)^4 - (1+x)^3 (1+2x)^4$ is $8a+1$

9. $\frac{(1+x)^{20}}{x^{10}} \Rightarrow$ independent term is ${}^{20}C_{10}$
10. ${}^nC_3 (px)^{n-3} \frac{1}{x^3} = \frac{5}{2}$
11. Use the formula $r = \frac{np}{p+q}$
12. Middle Terms are $n+1, n+2$
13. $\frac{T_{r+1}}{T_r}$
14. ${}^nC_{r-1} : {}^nC_r = 4:1$
15. Use the formula $\frac{(n+1)|x|}{1+|x|}$
16. $T_{17} = T_{18}$ simplify
17. Multiply and divide with $1-x$ and simplify
18. $T_{21} = T_{22}$
19. Put $x=1$
20. Expand $(1-x)^8$ and pick up coeff. x^{10}
21. $2[{}^{100}C_0 x^{100} + {}^{100}C_2 x^{98} a^2 \dots\dots\dots]$
22. $\left[\frac{124}{L.C.M \text{ of } (2,4)} \right] + 1$
23. Substitute a positive integer for n and verify
24. Substtute n and Verify the options
25. $c_0 x + c_1 \frac{x^2}{2} + \dots + c_n \frac{x^{n+1}}{n+1} = \frac{(1+x)^{n+1} - 1}{n+1}$
put $x=2$
26. Substitute a positive integer for n and verify
27. Substitute a positive integer for n and verify
28. $(1+a)^{2n} = c_0 + c_1 a + c_2 a^2 + \dots c_{2n} a^{2n}$
differentiate both sides w.r.t a
29. Differentiate with respect x both sides and then put $x=1$
30. Differentiate and put $x=-1$
31. $2^n = 4096$, Find and $n_{c_{n/2}}$
32. $4PQ = (P+Q)^2 - (P-Q)^2$
33. $\left| \frac{7x}{4} \right| < 1$

34. Coefficient of x^2 in $\frac{1}{2} \left(1 - \frac{3x}{2} \right)^{-1}$
35. $(1+x)^2 (1-x)^{-3}$
36. Split into partial fractions and use infinite series formula
37. $(1+x)^{-4}$
38. $4 \left(1 + \frac{x^2}{16} \right)^{1/2} - 3 \left(1 + \frac{x^2}{9} \right)^{1/2}$
- 39 to 43. $(1-x)^{-p/q}$
44. $(1.01)^{1000000} = (1+0.01)^{1000000}$
45. $\frac{f(1)+f(-1)}{2}$
46. $\frac{c}{l} < 1$
47. $(n+1)(n+2) = 72$
48. Apply multi nomial

 LEVEL-I (H.W)

PROBLEMS BASED ON COEFFICIENT OF x^k

1. The coefficient of x^{30} in $\left(3x^2 + \frac{2}{3x^2} \right)^{15}$ is
 1) ${}^{15}C_2 \cdot 3^8 \cdot 2^7$ 2) ${}^{15}C_1 \cdot 3^7 \cdot 2^8$
 3) 3^{15} 4) $3^{14} \cdot 2^2$
2. The coefficient of y in $\left(y^2 + \frac{c}{y} \right)^5$ is
 1) $20c$ 2) $10c$ 3) $10c^3$ 4) $20c^2$
3. The greatest binomial coefficient in the expansion of $\left(\frac{x^{\frac{3}{2}} \cdot y}{2} + \frac{2}{x \cdot y^{\frac{3}{2}}} \right)^{12}$ is
 1) ${}^{12}C_5$ 2) ${}^{12}C_6$ 3) ${}^{12}C_5 \cdot 2^2$ 4) ${}^{12}C_6 \cdot 2^3$
4. If the coefficients of $(2r+4)$ th term and $(r-2)$ th term in the expansion of $(1+x)^{18}$ are equal, then r =
 1) 4 2) 5 3) 6 4) 7

5. Coefficient of x^5 in the expansion of

$$(1+x)^{10} \cdot \left(1 + \frac{1}{x}\right)^{20} \text{ is}$$

- 1) ${}^{30}C_5$ 2) ${}^{10}C_5$ 3) ${}^{20}C_5$ 4) ${}^{30}C_{20}$

6. The coefficient of x^{-17} in $\left(x^4 - \frac{1}{x^3}\right)^{15}$ is

- 1) $-{}^{15}C_{11}$ 2) ${}^{15}C_{11}$ 3) ${}^{15}C_5$ 4) $-{}^{15}C_5$

7. If the coefficients of $(3r)$ th and $(r+2)$ th terms in the expansion of $(1+x)^{2n}$ are equal ($r>1, n>2$), then

- 1) $n=2r$ 2) $n=3r$ 3) $n=2r+1$ 4) $n=4r$

INDEPENDENT TERM & MIDDLE TERM

8. The term independent of x in $\left(2x - \frac{1}{2x^2}\right)^{12}$ is

- 1) $-{}^{12}C_3 \cdot 2^6$ 2) $-{}^{12}C_5 \cdot 2^2$ 3) ${}^{12}C_6$ 4) ${}^{12}C_4 \cdot 2^4$

9. The term independent of x in $\left(2\sqrt{x} - \frac{3}{x^3}\right)^{20}$ is

- 1) ${}^{20}C_8 \cdot 2^8 \cdot 3^{12}$ 2) $-{}^{20}C_8 \cdot 2^8 \cdot 3^{12}$
3) ${}^{20}C_7 \cdot 2^7 \cdot 3^{13}$ 4) $-{}^{20}C_7 \cdot 2^7 \cdot 3^{13}$

10. The term independent of x in $\left(x^2 - \frac{1}{x}\right)^9$ is

- 1) 1 2) -1 3) 48 4) 84

11. The term independent of x in the expansion of

$$\left(\frac{\sqrt{x}}{2} - \frac{2}{x^2}\right)^{10} \text{ is ...}$$

- 1) $45/64$ 2) $64/45$ 3) $6/5$ 4) $4/5$

12. In $\left(\sqrt{x} - \frac{2}{x}\right)^{18}$, the term independent of x is

- 1) ${}^{18}C_6$ 2) $2 \cdot {}^{18}C_6$ 3) ${}^{18}C_6 \cdot 2^6$ 4) $16 \cdot {}^{18}C_6$

13. The middle term in the expansion of $(1+x)^{2n}$ is

- 1) ${}^{2n}C_n$ 2) ${}^{2n}C_{n-1} \cdot X^{n+1}$ 3) ${}^{2n}C_{n-1} \cdot X^{n-1}$ 4) ${}^{2n}C_n \cdot X^n$

PROBLEMS BASED ON RATIOS & NUMERICALLY GREATEST TERM

14. The ratio of the coefficient of x^{10} in $(1-x^2)^{10}$ and

$$\text{the term independent of } x \text{ in } \left(x - \frac{2}{x}\right)^{10} \text{ is}$$

- 1) $32 : 1$ 2) $-32 : 1$ 3) $-1 : 32$ 4) $1 : 32$

15. The ratio of the r th term and the $(r+1)$ th term in the expansion of $(1+x)^n$ is

- 1) $\frac{r}{(n-r+1)x}$ 2) $\frac{1}{(n-r+1)x}$
3) $\frac{r}{n-r+1}$ 4) $\frac{(n-r+1)x}{r}$

16. When $x=1, y=\frac{3}{2}$, numerically greatest term in the expansion of $(2x-3y)^{12}$ is

- 1) 6th 2) 8th 3) 10th 4) 12th

PROBLEMS BASED ON COEFFICIENT

17. If the first three terms in the expansion of $(1+ax)^n$ are 1, $8x$, $24x^2$ respectively, then $a=$

- 1) 1 2) 2 3) 4 4) 8

18. The coefficients of 2nd, 3rd and 4th terms in the expansion of $(1+x)^n$ are in A.P. then $n=$

- 1) 7 2) 8 3) 9 4) 14

19. In the binomial expansion of $\left(\sqrt{y} + \frac{1}{2\sqrt[4]{y}}\right)^8$, the terms in the expansion in which the power of y is a natural number are

- 1) T_1, T_5 2) T_1, T_2, T_3, T_4
3) T_2, T_4 4) T_1, T_3, T_5

20. In the expansion of $(1+x)^n$, the binomial coefficients of 3 consecutive terms are respectively 220, 495 and 792, then $n=$

- 1) 4 2) 8 3) 12 4) 16

21. The coefficients of 9th, 10th and 11th terms in the expansion of $(1+x)^n$ are in A.P. then $n=$

- 1) 7 2) 7 or 14 3) 14 4) 21

22. In the expansion of $(x+y)^n$, if the binomial coefficient of the third term is greater by 9 than that of the second term, then the sum of the binomial coefficients of the terms occupying the odd places is

- 1) 2^9 2) 2^6 3) 2^5 4) 2^8

23. The value of $(1.02)^4 + (0.98)^4$ correct to three decimal places is

- 1) 2.004 2) 2.005 3) 2.0064 4) 1.995

24. Coefficient of x^4 in $(1+x-2x^2)^6$ is

- 1) -60 2) -45 3) 45 4) 15

25. Sum of the third from the beginning and the third from the end of the binomial coefficients of the expansion of $(\sqrt[4]{3} + \sqrt[3]{4})^n$ is equal to 9900. Number of rational terms contained in the expansion is
1) 8 2) 9 3) 10 4) 11
26. The sum of the coefficients in the expansion of $(1+x-3x^2)^{2143}$ is
1) 0 2) 1 3) -1 4) $(-2)^{2143}$
27. Sum of the coefficients in the expansion of $(5x-4y)^n$ where n is a positive integer is
1) 1 2) 9^n 3) $(-1)^n$ 4) 5^n
28. If $(1+x-2x^2)^6 = \sum_{r=0}^{12} a_r \cdot x^r$, then $a_2 + a_4 + \dots + a_{12} =$
1) 64 2) 63 3) 32 4) 31

BINOMIAL COEFFICIENTS

29. ${}^{2n}C_2 + {}^{2n}C_4 + \dots + {}^{2n}C_{2n} =$
1) 2^{2n} 2) 2^{2n-1} 3) 2^{2n-1} 4) $2^{2n-1}-1$
30. $C_0 + 2.C_1 + 3.C_2 + \dots + (n+1).C_n =$
1) $(n+2)2^n$ 2) $n.2^n$ 3) $n.2^{n-1}$ 4) $(n+2)2^{n-1}$
31. $\frac{C_1}{C_0} + 2\frac{C_2}{C_1} + 3\frac{C_3}{C_2} + \dots + n\frac{C_n}{C_{n-1}} =$
1) $\frac{(n+1)(n+2)}{2}$ 2) $\frac{n(n+1)}{2}$
3) $\frac{n(n-1)}{2}$ 4) $\frac{n(n+2)}{2}$
32. The sum to $(n+1)$ terms of the series $\frac{C_0}{2} - \frac{C_1}{3} + \frac{C_2}{4} - \frac{C_3}{5} + \dots =$
1) $\frac{1}{n+1}$ 2) $\frac{1}{n+2}$
3) $\frac{1}{n(n+1)}$ 4) $\frac{1}{(n+1)(n+2)}$
33. $C_0 + 3.C_1 + 5.C_2 + \dots + (2n+1).C_n =$
1) $(n+1)2^n$ 2) $(2n+1)2^{n-1}$
3) $(2n+1)2^n$ 4) $(n+1)2^{n-1}$
34. $5.C_0 + 8.C_1 + 11.C_2 + \dots$ to $(n+1)$ terms =
1) $(3n+5)2^n$ 2) $(3n+5)2^{n-1}$
3) $(3n+10)2^n$ 4) $(3n+10)2^{n-1}$
35. ${}^5C_0 + 2.{}^5C_1 + 2^2.{}^5C_2 + 2^3.{}^5C_3 + 2^4.{}^5C_4 + 2^5.{}^5C_5 =$
1) 32 2) 243 3) 64 4) 729

BINOMIAL THEOREM FOR RATIONAL INDEX

36. In the expansion of $\frac{1}{1-3x}$, coefficient of x^4 is
1) 81 2) 27 3) -27 4) 28
37. In the expansion of $(1+x+x^2+x^3+x^4)^{-2}$, coefficient of x^{10} is
1) 3 2) 5 3) 7 4) 9
38. In the expansion of $\frac{3-x}{(1-x)^2}$ coefficient of x^r is 9 then r =
1) 3 2) 4 3) 3.5 4) 4.5
39. If n is a positive integer, then the coefficient of x^n in the expansion of $\frac{(1+2x)^n}{1-x}$ is
1) $n.3^n$ 2) $(n-1)3^n$ 3) $(n+1)3^n$ 4) 3^n

INFINITE SERIES

40. $1+n\left(\frac{2n}{1+n}\right) + \frac{n(n+1)}{2!}\left(\frac{2n}{1+n}\right)^2 + \dots \infty =$
1) $\left(\frac{1-n}{1+n}\right)^n$ 2) $\left(\frac{1+n}{1-n}\right)^n$ 3) $\left(\frac{1+3n}{1+n}\right)^n$ 4) $\left(\frac{1+n}{1+3n}\right)^n$
41. Neglecting x^n for $n \geq 2$, value of $(1-7x)^{\frac{1}{3}}(1+2x)^{\frac{3}{4}}$ is
1) $1 - \frac{5x}{6}$ 2) $1 + \frac{5x}{6}$ 3) $1 + \frac{23x}{6}$ 4) $1 - \frac{23x}{6}$
42. $1 + \frac{1}{4} + \frac{1.4}{4.8} + \frac{1.4.7}{4.8.12} + \dots \infty =$
1) $\left(\frac{1}{4}\right)^{\frac{1}{3}}$ 2) $\left(\frac{1}{2}\right)^{\frac{1}{3}}$ 3) $(2)^{\frac{1}{3}}$ 4) $(4)^{\frac{1}{3}}$
43. $\frac{3}{6} + \frac{3.5}{6.9} + \frac{3.5.7}{6.9.12} + \dots \infty =$
1) $3\sqrt{3}$ 2) $3\sqrt{3} - \frac{4}{3}$ 3) $3\sqrt{3} - 4$ 4) $2\sqrt{3}$
44. If $1 + \frac{2}{x} + \frac{3}{x^2} + \frac{4}{x^3} + \dots \infty = \frac{1}{2}$, then x =
1) $-1 + \sqrt{2}$ 2) $-1 - \sqrt{2}$ 3) $1 + \sqrt{2}$ 4) $1 - \sqrt{2}$

45. $1 + \frac{1}{2} \cdot \frac{3}{5} + \frac{1 \cdot 3}{2 \cdot 4} \left(\frac{3}{5}\right)^2 + \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6} \left(\frac{3}{5}\right)^3 + \dots \infty =$

1) $\sqrt{\frac{5}{2}}$ 2) $\sqrt{\frac{2}{5}}$ 3) $\frac{1}{2}\sqrt{\frac{5}{2}}$ 4) $\sqrt{\frac{5}{3}}$

46. $2 + \frac{5}{2! \cdot 3} + \frac{5 \cdot 7}{3! \cdot 3^2} + \dots \infty =$

1) $3\sqrt{3} - 1$ 2) $\sqrt{3}$ 3) $3\sqrt{3} + 1$ 4) $3\sqrt{3}$

BINOMIAL THEOREM APPLICATIONS

47. If 17^{th} and 18^{th} terms of the expansion of $(2+a)^{50}$ are equal then value of a is

1) 1 2) 2 3) 3 4) 4

48. $c_0^2 + c_1^2 + c_2^2 + \dots + c_{25}^2 =$ (where $nc_r = 25c_r$)

1) $49C_{50}$ 2) $49C_{25}$ 3) $50C_{25}$ 4) $39C_{40}$

49. let p_n denotes product of binomial

coefficients in $(1+x)^n$ then $\frac{p_{n+1}}{p_n} =$

1) $\frac{(n+1)^n}{n!}$ 2) $\frac{(n+1)^{2n}}{n!}$ 3) $\frac{(n+1)^{2n}}{(n!)^2}$ 4) $\frac{(n+1)^2}{(n!)^2}$

50. If $9^7 + 7^9$ is divisible by 2^n then the greatest value of n , where $n \in \mathbb{N}$

1) 6 2) 31 3) 5 4) 7

51. The sum of $\sum_{r=0}^n {}^{n+r}C_r$

1) ${}^{2n}C_n$ 2) ${}^{2n+1}C_n$ 3) ${}^{2n+1}C_{n+1}$ 4) $3^{n+1}C_{n+1}$

LEVEL-I (H.W) - KEY

- 1) 3 2) 3 3) 2 4) 3 5) 1 6) 1 7) 1
8) 4 9) 1 10) 4 11) 1 12) 3 13) 4 14) 4
15) 1 16) 3 17) 2 18) 1 19) 1 20) 3 21) 3
22) 3 23) 2 24) 2 25) 2 26) 3 27) 1 28) 4
29) 4 30) 4 31) 2 32) 4 33) 1 34) 4 35) 2
36) 1 37) 1 38) 1 39) 4 40) 2 41) 4 42) 4
43) 3 44) 2 45) 1 46) 4 47) 1 48) 3 49) 1
50) 1 51) 3

LEVEL-I (H.W) - HINTS

1. $r = \frac{np-m}{p+q}$

2. $r = \frac{np-m}{p+q}$

3. n is even use the formula ${}^nC_{n/2}$

4. ${}^{18}C_{2r+3} = {}^{18}C_{r-3}$

5. Coefficient of x^5 in $(1+x)^{10} \left(1+\frac{1}{x}\right)^{20} =$

Coefficient of x^{25} in $(1+x)^{30}$

6. $r = \frac{np-m}{p+q}$

7. ${}^{2n}C_{3r-1} = {}^{2n}C_{r+1}$

8 to 13. Use the formula $r = \frac{np}{p+q}$

13. Middle term $= \frac{n}{2} + 1$

14. $\frac{T_6}{T_6} = \frac{(-1)}{(-2)^5} = \frac{1}{32}$

15. $\frac{T_r}{T_{r+1}}$

16. Use the formula $\frac{(n+1)|x|}{1+|x|}$

17. $T_1 = 1, T_2 = 8x, T_3 = 24x^2$

18. $(n-2r)^2 = n+2$

19. $T_{r+1} = {}^8C_r (\sqrt{y})^{8-r} \left(\frac{1}{2\sqrt[4]{y}}\right)^r$ verify for which

$\frac{8-r}{2} - \frac{r}{4}$ is an interger

20. Frame the equations and solve

21. $(n-2r)^2 = n+2$

22. $n_{c_2} - n_{c_1} = 9$

23. $(1+.02)^4 + (1-0.02)^4 =$

24. Factorise and expand
25. $T_3 + T_{n-3+2} = 9900$
26. Put $x = 1$
27. Put $x = y = 1$
28. Put $x = 1$; $x = -1$, and then add
29. Substitute a positive integer for n and verify
30. Substitute n and Verify the options
31. $\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r}$
32. Substitute a positive integer for n and verify
33. Substitute n and Verify the options
34. Substitute n and Verify the options
35. Simplify
36. Coefficient of x^4 in $(1-3x)^{-1}$
37. $(1-x)^2(1-x^5)^{-2}$ Expand and pick up the coefficient of x^{10}
38. $(3-x)(1-x)^{-2}$
39. $(1+2x)^n(1-x)^{-1}$
40. $(1+x)^{-p/q}$
41. Expand and neglect x^2 , higher terms.
42. $(1-x)^{-p/q}$
43. Adjust the series and apply the formula $(1-x)^{-\frac{p}{q}}$
44. Put $\frac{1}{x} = t$, $(1-t)^{-2} = \frac{1}{2}$
45. $(1-x)^{-p/q}$
46. Adjust the series and apply the formula $(1-x)^{-\frac{p}{q}}$
47. Equate the terms
48. $c_0^2 + c_1^2 + \dots + c_n^2 = 2nc_n$ $c_0^2 + c_1^2 + \dots + c_n^2 = 2nc_n$
49. Simplify Nr & Dr
50. $(8+1)^7 + (8-1)^9$ is divisible by 8^2
51. Simplify



LEVEL-II - CW

PROBLEMS BASED ON COEFFICIENT OF x^k , INDEPENDENT TERM AND MIDDLE TERM

1. Coefficient of x in the expansion of $(1-2x^3+3x^5)\left(1+\frac{1}{x}\right)^8$ is
1) 154 2) 164 3) 146 4) 156
2. The middle term in the expansion of $(1-3x+3x^2-x^3)^{2n}$ is
1) ${}^{6n}C_{3n}(-x)^{3n}$ 2) ${}^{6n}C_{2n}(-x)^{2n+1}$
3) ${}^{4n}C_{3n}(-x)^{3n}$ 4) ${}^{6n}C_{3n}(-x)^{3n-1}$
3. In the expansion of $(1+x)^n$, the 5th term is 4 times the 4th term and the 4th term is 6 times the 3rd term. then $n =$
1) 9 2) 10 3) 11 4) 12

PROBLEMS BASED ON COEFFICIENTS

4. In the expansion $(1+x)^m(1-x)^n$, the coefficients of x and x^2 are 3 and -6 respectively, then m is
1) 6 2) 9 3) 12 4) 24
5. If $(1+kx)^{10} = a_0 + a_1x + a_2x^2 + \dots + a_{10}x^{10}$ and $a_2 + \frac{7}{5}a_1 + 1 = 0$ then $k =$
1) $-1/5, -1/9$ 2) $1/5, 1/9$
3) $-1/5, -1/7$ 4) $1/5, -1/9$

PROBLEMS BASED ON BINOMIAL COEFFICIENTS

6. $C_1 + 2.5C_2 + 3.5^2C_3 + \dots =$
1) $n.6^{n-1}$ 2) 6^{n-1} 3) 6^n 4) $n.6^n$
7. $\frac{(C_0 + C_1)(C_1 + C_2)(C_2 + C_3) \dots (C_{n-1} + C_n)}{C_0C_1C_2 \dots C_n} =$
1) $\frac{(n+1)^n}{n}$ 2) $\frac{n+1}{\angle n}$ 3) $\frac{(n+1)^{n-1}}{\angle n}$ 4) $\frac{(n+1)^n}{\angle n}$

8. $\sum_{r=2}^n (5r-3)C_r =$

- 1) $(5n+6)2^{n-1} - 2n+2$ 2) $(5n+6)2^{n-1} - 2n+3$
3) $(5n-6)2^{n-1} - 2n+2$ 4) $(5n-6)2^{n-1} - 2n+3$

PROBLEMS BASED ON SUM OF THE COEFFICIENTS

9. The sum of the coefficients of even powers of x in the expansion $(1-x+x^2-x^3)^5$ is

- 1) 512 2) 0 3) -512 4) 510

10. The sum of the coefficients of middle terms in the expansion of $(1+x)^{2n-1}$

- 1) $(2n)!$ 2) $\frac{(2n)!}{n!}$ 3) $\frac{(2n)!}{(n!)^2}$ 4) $\frac{(2n-1)!}{n!}$

11. The sum of the coefficients of the first 10 terms in the expansion of $(1-x)^{-3}$ is

- 1) 220 2) 286 3) 120 4) 150

12. If $(1-x-x^2)^{20} = \sum_{r=0}^{40} a_r \cdot x^r$, then

$a_1 + 3a_3 + 5a_5 + \dots + 39a_{39} =$
1) 40 2) -40 3) 80 4) -80

PROBLEMS BASED ON INFINITE SERIES

13. The coefficient of x^{24} in the expansion of

$(1+3x+6x^2+10x^3+\dots+\infty)^{2/3} =$
1) 300 2) 250 3) 25 4) 205

14. If S_n denotes the sum of first 'n' natural numbers, then

$S_1 + S_2x + S_3x^2 + \dots + S_nx^{n-1} + \dots =$
1) $(1-x)^{-1}$ 2) $(1-x)^{-2}$ 3) $(1-x)^{-3}$ 4) $(1-x)^{-4}$

15. If $\frac{1}{1-2x+x^2} = 1 + b_1x + b_2x^2 + b_3x^3 + \dots$

$|x| < 1$, then the value of b_1 is
1) 3 2) 2 3) 4 4) 1

16. If $y = 2x + 3x^2 + 4x^3 + \dots$, then

$\frac{y}{2} - \frac{1.3}{2!} \left(\frac{y}{2}\right)^2 + \frac{1.3.5}{3!} \left(\frac{y}{2}\right)^3 - \dots =$
1) $x+2$ 2) x 3) $x-2$ 4) $x+4$

17. $1 + \frac{2.1}{3.2} + \frac{2.5.1}{3.6.2^2} + \frac{2.5.8}{3.6.9} \cdot \frac{1}{2^3} + \dots + \infty =$

- 1) 0.4 (nearly) 2) 0.3 3) $\sqrt{3}$ 4) $\sqrt{3}-1$

18. The greatest integer less than or equal to $(2+\sqrt{3})^6$ is

- 1) 2702 2) 2701 3) 2700 4) 2699

19. $(2-\sqrt{5})^6 + (2+\sqrt{5})^6 =$

- 1) 1264 2) 1964 3) 2889 4) 5778

20. The number of terms in the expansion of $[(a+4b)^3 + (a-4b)^3]^2$ are

- 1) 6 2) 8 3) 7 4) 3

21. The coefficient of x^{17} in the expansion of $(x-1)(x-2)(x-3)\dots(x-18)$ is

- 1) 164 2) -171 3) 194 4) 221

22. If $(1+x+x^2)^n = \sum_{r=0}^{2n} a_r \cdot x^r$ then

$a_1 - 2a_2 + 3a_3 - \dots - 2na_{2n} = \dots$
1) 0 2) 1 3) n 4) $-n$

LEVEL-II (C.W) - KEY

- 1) 1 2) 1 3) 3 4) 3 5) 1 6) 1 7) 4
8) 4 9) 1 10) 3 11) 1 12) 1 13) 3 14) 3
15) 2 16) 2 17) 1 18) 2 19) 4 20) 4 21) 2
22) 4

LEVEL-II (C.W) - KEY

- Expand using B.T and pickup coefficient of x
- $[(1-x)^3]^{2n} = (1-x)^{6n}$
middle term $= {}^{6n}C_{3n}(-x)^{3n}$
- $T_5 = 4.T_4$; $T_4 = 6.T_3$
 ${}^nC_4 \cdot x^4 = 4 \cdot {}^nC_3 \cdot x^3 = 6 \cdot {}^nC_2 \cdot x^2$
 $\therefore n = 11$
- Expand, multiply and pick relevant coefficients
- Expand, compare the coefficients
- Put $n = 2$, 1st option is satisfied
- substitute a positive integer for n and verify

8. $5 \sum_{r=2}^n r.C_r - 3 \sum_{r=2}^n C_r$
 $= 5[n.2^{n-1} - n] - 3[2^n - n - 1]$
 $= 2^{n-1}(5n - 6) - 2n + 3$
9. $\frac{f(1)+f(-1)}{2}$
10. $(2n-1)C_n + (2n-1)C_{n-1} = 2^n C_n$
11. The sum of coefficients of the first r terms in $(1-x)^{-n}$ is $n+r-1 C_n$
12. $\frac{f'(-1)+f'(1)}{2}$
13. $\left[(1-x)^{-3}\right]^{\frac{2}{3}} = (1-x)^{-2}$
 Formula: coefficient of x^r in $(1-x)^{-n}$ is $(r+n-1)C_r$
14. $1 + (1+2)x + (1+2+3)x^2 + \dots$
 $= 1 + 3x + 6x^2 + \dots = (1-x)^{-3}$
15. $1 + b_1x + b_2x^2 + \dots = (1-x)^{-2}$
 $b_1 = 2$
16. $1 + y = (1-x)^{-2}$
 $(1-x) = (1+y)^{-\frac{1}{2}} \Rightarrow x = 1 - (1+y)^{-\frac{1}{2}}$
17. $n = \frac{2}{3}, x = \frac{1}{2}$
 $\left(1 - \frac{1}{2}\right)^{-2/3} = \sqrt[3]{4} = 0.4(\text{nearly})$
18. $(2 + \sqrt{3})^6 + (2 - \sqrt{3})^6 = 2({}^6C_0 2^6 + \dots)$
19. $2({}^6C_0 2^6 + {}^6C_2 2^4 5 + {}^6C_4 2^2 5^2 + {}^6C_6 5^3)$
20. $[2(a^3 + 3.a.16b^2)]^2$
21. $-1 - 2 - 3 - \dots - 18 = -\sum 18 = -171$
22. $(1 + x + x^2)^n = \sum_{r=0}^{2n} a_r x^r$
 differentiate above and put $x = -1$



LEVEL-II - HW

PROBLEMS BASED ON COEFFICIENT OF x^k , GENERAL TERM

1. The term independent of x in the expansion

$$\left(1 + 2x + \frac{2}{x}\right)^3 \text{ is}$$

- 1) 12 2) 18 3) 36 4) 25

2. If the 5th term of $\left(\frac{q}{2x} - px\right)^8$ is 1120 and

$$p + q = 5, p > q \text{ then } p =$$

- 1) 3 2) 6 3) 4 4) 7

3. The coefficient of x^{53} in the expansion of

$$\sum_{m=0}^{100} {}^{100}C_m (x-3)^{100-m} \cdot 2^m \text{ is}$$

- 1) $C(100, 53)$ 2) $-C(100, 43)$
 3) $-C(100, 53)$ 4) ${}^{100}C_{50}$

BINOMIAL COEFFICIENTS

4. If $C_0 + C_1 + C_2 + \dots + C_n = 128$ then

$$C_0 - \frac{C_1}{2} + \frac{C_2}{3} - \frac{C_3}{4} + \dots =$$

- 1) 0 2) 8 3) $1/8$ 4) $7/8$

5. $(n+1)C_1 + (n+1)C_2 + (n+1)C_3 + \dots + (n+1)C_n =$

- 1) $2(2^n + 1)$ 2) $2(2^n - 1)$
 3) 2^{n+1} 4) $(2^{n+1} - 1)$

6. $\int_0^1 (1-x^3)^n dx =$

- 1) $C_0 - \frac{C_1}{4} + \frac{C_2}{7} - \frac{C_3}{10} + \dots + (-1)^n \frac{C_n}{3n+1}$
 2) $C_0 + \frac{C_1}{4} + \frac{C_2}{7} + \frac{C_3}{10} + \dots + \frac{C_n}{3n+1}$
 3) $C_0 + \frac{C_1}{4} + \frac{C_2}{7} + \frac{C_3}{10} + \dots + \frac{C_{n-1}}{3n-2}$
 4) $C_0 C_1 + C_1 C_2 + C_2 C_3 + \dots + C_{(n-1)} C_n$

PROBLEMS BASED ON SUM OF THE COEFFICIENTS

7. The sum of the coefficients of odd powers of x in the expansion $(1+x+x^2+x^3)^5$ is
1) 520 2) 525 3) 576 4) 512
8. If the sum of the coefficients in the expansion of $(a^2x^2 - 2ax + 1)^{51}$ vanishes, then the value of a is
1) 2 2) -1 3) 1 4) -2
9. The sum of the binomial coefficients of the 3^{rd} , 4^{th} terms from the beginning and from the end of $(a+x)^n$ is 440 then $n =$
1) 10 2) 11 3) 12 4) 13
10. Using the binomial expansion upto 3 terms the approximate value of $(8.8)^{1/3}$ can be shown to be
1) $2 + \frac{67}{900}$ 2) $2 + \frac{107}{900}$ 3) $2 + \frac{58}{900}$ 4) $2 + \frac{47}{900}$
11. If cubes and higher powers of x are negligible then one can write $\sqrt{\frac{1-x}{1+x}} = 1 + Ax + Bx^2$, then $A =$
1) -1 2) 1 3) 0 4) 2
12. The coefficient of x^4 in the expansion of $\frac{3x-8}{4-4x+x^2}$ is
1) -1/4 2) -4 3) 4 4) 1/4
13. The number of non zero terms in $(x+a)^{75} + (x-a)^{75}$
1) 38 2) 76 3) 34 4) 32
14. $(\sqrt{2}+1)^6 + (\sqrt{2}-1)^6 =$
1) 99 2) 98 3) 196 4) 198
15. Let $R = (5\sqrt{5}+11)^{2n+1}$, $f = R - [R]$, then $Rf =$
1) 1 2) 2^n 3) 2^{2n} 4) 4^{2n+1}
16. If $(6 + \sqrt{35})^n = I + F$ when I is odd and $0 < F < 1$, then $(I+F)(1-F) =$
1) 1 2) 1/2 3) 2 4) 4
17. $(x-1)^4 + 4(x-1)^3 + 6(x-1)^2 + 4(x-1) + 1 =$
1) x^4 2) x^3 3) x^2 4) 1

18. Let $(1+\sqrt{2})^n = x_n + y_n\sqrt{2}$ where x_n, y_n are integers, then

$$1) x_n^2 - 2y_n^2 = (-1)^n \quad 2) x_n + 2y_n - x_{n+1} = 3$$

$$3) x_n^2 - 2y_n^2 = 1 \quad 4) x_{n+1} - x_n - 2y_n = 1$$

LEVEL-II (H.W) - KEY

- 1) 4 2) 3 3) 3 4) 3 5) 2 6) 1 7) 4
- 8) 3 9) 2 10) 3 11) 1 12) 1 13) 1 14) 4
- 15) 4 16) 1 17) 1 18) 1

LEVEL-II (H.W) - HINTS

1. Expand $\left(1 + 2\left(x + \frac{1}{x}\right)\right)^3$
2. $T_5 = 1120$
3. Coefficient of x^{53} is $^{100}C_{53} (-1)^{53}$
4. $2^n = 128 \Rightarrow n = 7$
5. $^{(n+1)}C_0 + ^{(n+1)}C_1 + \dots + ^{(n+1)}C_n + \dots + ^{(n+1)}C_{n+1} = 2^{n+1}$
6. Put $n = 1; \int_0^1 (1-x^3) dx = \left[x - \frac{x^4}{4}\right]_0^1 = \frac{3}{4}$
7. $\frac{f(1)-f(-1)}{2}$
8. $f(1) = 0 \Rightarrow a^2 - 2a + 1 = 0 \Rightarrow a = 1$
9. $^nC_2 + ^nC_3 + ^nC_{n-2} + ^nC_{n-3} = 440$
 $2(^nC_2 + ^nC_3) = 440, n = 11$
10. $2[1+.1]^{1/3}$
11. $\left(\frac{1-x}{1+x}\right)^{\frac{1}{2}} = 1 - \frac{1}{2}\left(\frac{2x}{1+x}\right) + \dots$
 $\therefore A = -1$
12. $\frac{(3x-8)}{(2-x)^2} = \frac{(3x-8)}{4} \left(1 - \frac{x}{2}\right)^{-2}$
13. $\frac{75+1}{2} (\because n \text{ is odd})$

14. $2\left({}^6C_0(\sqrt{2})^6 \dots\right)$

15. $\left[(5\sqrt{5})^2 - (11)^2\right]^{2n+1}$

16. $(6 + \sqrt{35})^n (6 - \sqrt{35})^n = 1$

17. $(x - 1 + 1)^4 = x^4$

18. $(1 + \sqrt{2})^n = x_n + y_n \sqrt{2}$

$(1 - \sqrt{2})^n = x_n - y_n \sqrt{2}$

$(1 + \sqrt{2})^n (1 - \sqrt{2})^n = x_n^2 - 2y_n^2$

$(-1)^n = x_n^2 - 2y_n^2$



LEVEL-III

1. The expansion

$\left[x + (x^3 - 1)^{1/2}\right]^5 + \left[x - (x^3 - 1)^{1/2}\right]^5$ is a

polynomial of degree

- 1) 8 2) 7 3) 6 4) 5

2. If $x = (99)^{50} + (100)^{50}$ and $y = (101)^{50}$, then

- 1) $x > y$ 2) $x < y$ 3) $x = y$ 4) $xy = 1$

3. If the coefficients of x^{39} and x^{40} are equal in the expansion of $(p + qx)^{49}$. then the possible values of p and q are

- 1) 1, 5 2) 1, 4 3) 1, 3 4) 2, 7

4. If $x + y = 1$, then $\sum_{r=0}^n r \cdot {}^nC_r x^r y^{n-r} =$

- 1) 1 2) n 3) nx 4) ny

5. If x is nearly equal to 1, then $\frac{mx^m - nx^n}{m - n} =$

- 1) x^{m+n} 2) x^{m-n} 3) x^m 4) x^n

6. If p and q are the coefficients of x^n in $(1 + x)^{2n}$

and $(1 - 4x)^{-1/2}$, $|x| < \frac{1}{4}$ then

- 1) $p = q$ 2) $p = 2q$ 3) $q = 2p$ 4) $p + q = {}^{2n}C_n$

7. If $(1 + x + x^2)^n = a_0 + a_1x + a_2x^2 + \dots + a_{2n}x^{2n}$

then $a_0 + a_3 + a_6 + \dots =$

- 1) 3^n 2) 3^{n-1} 3) 3^{n-2} 4) 1

8. If m and n are +ve integers and $m > n$ and if $(1 + x)^{m+n}(1 - x)^{m-n}$ is expanded as a polynomial in x, then the coefficient of x^2 is

- 1) $2m^2 - n$ 2) $2n^2 - m$ 3) $2m^2 + n$ 4) $2n^2 + m$

9. $9^{11} + 11^9$ is divisible by

- 1) 7 2) 8 3) 9 4) 10

10. For $n \in \mathbb{N}$; $(1 + x)^n - nx - 1$ is divisible by

- 1) 2 2) x 3) x^2 4) x^3

11. If $t_0, t_1, t_2, \dots, t_n$ are the consecutive terms in the expansion $(x + a)^n$ then

$(t_0 - t_2 + t_4 - t_6 + \dots)^2 + (t_1 - t_3 + t_5 - \dots)^2 =$

- 1) $x^2 + a^2$ 2) $(x^2 + a^2)^n$
3) $x^2 - a^2$ 4) $(x^2 - a^2)^n$

12. Coefficient of x^{50} in

$(1 + x)^{1000} + 2x(1 + x)^{999} + 3x^2(1 + x)^{998} + \dots$ is

- 1) ${}^{1000}C_{50}$ 2) ${}^{1001}C_{50}$ 3) ${}^{1002}C_{50}$ 4) ${}^{1002}C_{49}$

13. A positive integer which is just greater than $(1 + 0.0001)^{10000}$ is

- 1) 3 2) 4 3) 5 4) 6

14. If n is an integer lying between 0 and 21, then the least value of $n!(21 - n)!$ is

- 1) $1! \cdot 20!$ 2) $11! \cdot 10!$ 3) $9! \cdot 12!$ 4) $2! \cdot 19!$

15. Coefficient of x^5 in the expansion of

$\frac{3x}{(x-1)^2(x+2)}$ is

- 1) $\frac{171}{32}$ 2) $\frac{171}{64}$ 3) $\frac{57}{32}$ 4) $\frac{57}{16}$

16. Coefficient of x^n in $\frac{(1+x)(1+2x)(1+3x)}{(1-x)(1-2x)(1-3x)}$ is

- 1) $12 - 30 \cdot 2^n - 20 \cdot 3^n$ 2) $12 - 30 \cdot 2^n + 20 \cdot 3^n$
3) $12 + 30 \cdot 2^n + 20 \cdot 3^n$ 4) $12 + 30 \cdot 2^n - 20 \cdot 3^n$

17. The coefficient of x^9 in $(x+2)(x+4)(x+8)\dots(x+1024)$ is
1) 2046 2) 1023 3) 55 4) 0
18. If $S_n = \sum_{r=0}^n \frac{1}{{}^nC_r}$ and $t_n = \sum_{r=0}^n \frac{r}{{}^nC_r}$, then $\frac{t_n}{S_n} =$
1) $\frac{1}{2}n$ 2) $\frac{2n-1}{2}$ 3) $n-1$ 4) $\frac{1}{2}n-1$
19. If $|x| < \frac{1}{2}$, then the coefficient of x^r in the expansion of $\frac{1+2x}{(1-2x)^2}$ is
1) $r \cdot 2^r$ 2) $(2r-1)2^r$ 3) $r \cdot 2^{2r+1}$ 4) $(2r+1)2^r$
20. For natural numbers m, n if $(1-y)^m(1+y)^n = 1 + a_1y + a_2y^2 + \dots$, and $a_1 = a_2 = 10$, then (m, n) is
1) (45, 35) 2) (35, 45) 3) (20, 45) 4) (35, 20)
21. $\sum_{r=0}^n (-1)^r \cdot {}^nC_r \frac{1+r \cdot \log_e 10}{(1+\log_e 10^n)^r} =$
1) 1 2) -1 3) n 4) 0
22. In the expansion of $(1+x)^n \cdot (1+y)^n \cdot (1+z)^n$, the sum of the coefficients of the terms of degree r is
1) $({}^nC_r)^3$ 2) ${}^{3n}C_r$ 3) $3 \cdot {}^nC_r$ 4) ${}^nC_{3r}$
23. The coefficient of x^n in the polynomial $(x+{}^nC_0)(x+3 \cdot {}^nC_1)(x+5 \cdot {}^nC_2)\dots[x+(2n+1) \cdot {}^nC_n]$ is
1) $n \cdot 2^n$ 2) $n \cdot 2^{n+1}$ 3) $(n+1)2^n$ 4) $n2^n+1$
24. Coefficient of x^{2m+1} in the expansion of $\frac{1}{(1+x)(1+x^2)(1+x^4)(1+x^8)\dots(1+x^{2^m})}$ ($|x| < 1$) is
1) 3 2) 2 3) 1 4) 0
25. If $n > 2$ and $C_r = {}^nC_r$, then $1^2 \cdot C_0 - 2^2 \cdot C_1 + 3^2 \cdot C_2 - \dots =$
1) 0 2) $(-1)^n$ 3) n 4) $-n$

26. If $x = (2 + \sqrt{3})^n$, $n \in N$ and $f = x - [x]$, then $\frac{f^2}{1-f}$ is
1) an irrational number
2) a non integer rational number
3) an odd number 4) an even number
27. The last four digits of the natural number 3^{100} are
1) 7231 2) 1231 3) 3451 4) 2001
28. If $C_r = {}^{30}C_r$, then $C_0 + C_4 + C_8 + \dots + C_{28} =$
1) 2^{28} 2) 2^{29} 3) 2^{30} 4) 2^{15}
29. If $C_r = {}^{32}C_r$, then $\sum_{r=0}^5 C_{6r} =$
1) $\frac{1}{6}(2^{32} - 3^{16} - 1)$ 2) $\frac{1}{6}(2^{32} + 3^{16} - 1)$
3) $\frac{1}{6}(2^{32} - 3^{16} + 1)$ 4) $\frac{1}{6}(2^{32} + 3^{16})$
30. The coefficient of x^8 in the expansion of $\left(1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \frac{x^8}{8!}\right)^2$ is
1) $\frac{1}{315}$ 2) $\frac{2}{315}$ 3) $\frac{1}{105}$ 4) $\frac{1}{210}$
31. The remainder when 23^{23} is divided by 53 is
1) 17 2) 21 3) 30 4) 43
32. If $2^{2006} - 2006$ divided by 7, the remainder is
1) 0 2) 1 3) 2 4) 4
33. The number of rational terms in the expansion of $(1 + \sqrt{2} + \sqrt[3]{3})^6$ is
1) 6 2) 7 3) 3 4) 8
34. ${}^{404}C_4 - {}^{303}C_4 \cdot {}^4C_1 + {}^{202}C_4 \cdot {}^4C_2 - {}^{101}C_4 \cdot {}^4C_3 =$
1) $(101)^3$ 2) $(101)^4$ 3) $(202)^3$ 4) $(202)^4$
35. If $\{x\}$ denotes fractional part of x then $\left\{\frac{2^{2003}}{17}\right\} =$
1) $\frac{2}{17}$ 2) $\frac{4}{17}$ 3) $\frac{6}{17}$ 4) $\frac{8}{17}$

36. If $(3 + x^{2008} + x^{2009})^{2010}$
 $= a_0 + a_1x + a_2x^2 + \dots + a_nx^n$
 then $a_0 - \frac{a_1}{2} - \frac{a_2}{2} + a_3 - \frac{a_4}{2} - \frac{a_5}{2} + a_6, \dots$
 1) 2^{2010} 2) 2^{2011} 3) 2^{2012} 4) 2^{2013}

LEVEL-III - KEY

- 1) 2 2) 2 3) 2 4) 3 5) 1 6) 1 7) 2
 8) 2 9) 4 10) 3 11) 2 12) 3 13) 1 14) 2
 15) 1 16) 2 17) 1 18) 1 19) 4 20) 2 21) 4
 22) 2 23) 3 24) 3 25) 1 26) 4 27) 4 28) 1
 29) 1 30) 1 31) 3 32) 1 33) 2 34) 2 35) 4
 36) 1

LEVEL-III - HINTS

- Expand using the formula $(a + \sqrt{b})^n + (a - \sqrt{b})^n$
- $(101)^{50} - (99)^{50} = (100 + 1)^{50} - (100 - 1)^{50}$
- ${}^{49}C_{39} p^{10} q^{39} = {}^{49}C_{40} p^9 q^{40} \frac{p}{10} = \frac{q}{40}$;
 $\therefore p, q = 1, 4$
- $\sum_{r=0}^n r \cdot {}^nC_r x^r \cdot y^{n-r} = \sum_{r=0}^n r \cdot \frac{n!}{(n-r)!r!} x^r \cdot y^{n-r}$
 $= nx \sum_{r=1}^n {}^{n-1}C_{r-1} x^{r-1} \cdot y^{n-r} = nx(y+x)^{n-1} = nx$
- Let $x = 1 + h$
 The given expression reduces to

$$= \frac{m(1+h)^m - n(1+h)^n}{m-n} \left(\begin{array}{l} \text{Neglecting higher} \\ \text{powers and simplifying} \end{array} \right)$$

$$= \frac{m(1+mh) - n(1+nh)}{m-n}$$

$$= 1 + (m+n)h = (1+h)^{m+n} = x^{m+n}$$
- Coefficient of $x^r = {}^{2n}C_r \therefore p = {}^{2n}C_n$

$$q = \frac{1.2.3.4.5.6 \dots (2r) 2^r}{2.4.6 \dots 2r(r!)} x^r = {}^{2n}C_r x^r$$
 $\therefore p = q$

- Put $x=1$, $x = \omega$, $x = \omega^2$ and add those then it is

$$\frac{f(1)}{3} = 3^{n-1}$$
- Put $m=2$ and $n=1$; $(1+x)^3(1-x)$
 Coefficient of $x^2 = 3 - 3 = 0$; option 2 is 0
- $(10-1)^{11} + (10+1)^9 \therefore$ divisible by 10
- $(1+x)^n = 1 + nx + \frac{n(n-1)}{2} x^2 + \dots$
- Expand $(x+ai)^n$ and $(x-ai)^n$ then multiply
- Take $(1+x)^{1000}$ as common, after simplification
 it becomes $(1+x)^{1002}$ coefficient of x^{50} is ${}^{1002}C_{50}$
- $10000C_0 + 10000C_1(0.0001) + \dots +$
 $10000C_{10000}(0.0001)^{10000} =$
- $n! \cdot (21-n)! \text{ is minimum is } \Rightarrow \frac{21!}{n!(21-n)!} \text{ is}$
 maximum.
- By using partial fractions
- By using partial fractions
- $2 + 4 + 8 + \dots + 1024$
- $t_n = \sum_{r=0}^n \frac{r}{{}^nC_r} = \sum_{r=0}^n \frac{n-r}{{}^nC_{n-r}} = \sum_{r=0}^n \frac{n-r}{{}^nC_r}$
 $= n \sum_{r=0}^n \frac{1}{{}^nC_r} - \sum_{r=0}^n \frac{r}{{}^nC_r} = n.s_n - t_n$
 $\Rightarrow 2t_n = ns_n \Rightarrow \frac{t_n}{s_n} = \frac{n}{2}$
- $(1+2x)(1-2x)^{-2}$
- $(1-y)^m(1+y)^n = 1 + a_1y + a_2y^2 + \dots,$
 $(1 + {}^nC_1 y + {}^nC_2 y^2 + \dots) = 1 + a_1y + a_2y^2 + \dots$
 $\Rightarrow a_1 = {}^nC_1 - {}^mC_1, a_2 = {}^mC_2 - {}^mC_1 {}^nC_1 + {}^nC_2$
 $\Rightarrow n-m=10, m^2+n^2-m-n-2mn=20$
 $\Rightarrow n-m=10, (m-n)^2 - (m+n)=20$
 $\Rightarrow n-m=10, (m+n)=80 \Rightarrow m=35, n=45.$
 $\Rightarrow (1 - {}^mC_1 y + {}^mC_2 y^2 - \dots)$
 $= n(n-1)x^2 + nx$

21. let $x = \log_e 10$

$$= \sum_{r=0}^n (-1)^r \cdot {}^n C_r \frac{(1+rx)}{(1+nx)^r}$$

$$= \sum_{r=0}^n (-1)^r \cdot {}^n C_r \frac{1}{(1+nx)^r} +$$

$$\sum_{r=0}^n (-1)^r \cdot \frac{n}{r} \cdot {}^{n-1} C_{r-1} \frac{rx}{(1+nx)^r}$$

Simplify above

22. Given expression contains $3n$ factors
Choose r brackets out of $3n$ brackets to get a term of degree r . Hence the coefficient is ${}^{3n} C_r$

23. Coefficient of

$$x^n = [{}^n C_0 + 3 \cdot {}^n C_1 + 5 \cdot {}^n C_2 + \dots + {}^{2n+1} C_n]$$

$$\frac{2n+1+1}{2} \cdot 2^n = (n+1)2^n$$

24. Multiplying the numerator and denominator by

$$1-x, \quad \frac{1-x}{1-x^{2^{m+1}}}$$

25. $C_0 - C_1x + C_2x^2 - \dots = (1-x)^n$

$$C_0x - C_1x^2 + C_2x^3 - \dots = x(1-x)^n$$

Differentiation yields

$$1 \cdot C_0 - 2C_1x + 3 \cdot C_2x^2 - \dots = (1-n+1x)(1-x)^{n-1}$$

multiply with x then differentiate and substitute $x = 1$

26. $(2+\sqrt{3})^n = [x] + f$ Let $(2-\sqrt{3})^n = f^1, 0 < f^1 < 1$

$$[x] + f + f^1 = 2[{}^n C_0 2^n + {}^n C_2 2^{n-2} 3 + \dots]$$

an even integer

$$\therefore f + f^1 = 1$$

$$\therefore [x] \text{ is an odd integer, } f^1 = 1 - f$$

$$\text{But } ([x] + f) f^1 = (2 + \sqrt{3})^n (2 - \sqrt{3})^n = 1$$

$$\therefore [x] = \frac{1}{f^1} - f = \frac{1}{1-f} - f = \frac{1-f+f^2}{1-f}$$

$$\therefore \frac{f^2}{1-f} = [x] - 1, \text{ an even integer}$$

27. $3^{100} = 9^{50} = (1-10)^{50}$

$$= 1 - 50 \cdot 10 + \frac{50 \cdot 49}{2} 10^2 - \frac{50 \cdot 49 \cdot 48}{6} 10^3 +$$

multiple of 10^4

$$= 1 - 500 + 122500 + \text{multiple of } 10^4$$

$$= 2001 + \text{multiple of } 10^4$$

28. $(1+x)^{30} = C_0 + C_1x + C_2x^2 + \dots + C_{30}x^{30}$

Set $x = 1, -1, i$ and $-i$ then add

$$C_0 + C_4 + C_8 + \dots + C_{28}$$

$$= \frac{1}{4} [2^{30} + (1+i)^{30} + (1-i)^{30}] = 2^{28}$$

29. $(1+x)^{32} = C_0 + C_1x + C_2x^2 + \dots + C_{32}x^{32}$

Set $x = 1, \text{cis } \frac{\pi}{3}, \text{cis } \frac{2\pi}{3}, -1, \text{cis } \frac{4\pi}{3}, \text{cis } \frac{5\pi}{3}$ in

turn and adding

we get

$$6[C_0 + C_6 + C_{12} + \dots + C_{30}]$$

$$= (1+1)^{32} + (1-1)^{32} + \left(\frac{3}{2} + i\frac{\sqrt{3}}{2}\right)^{32} + \left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)^{32} +$$

$$\left(\frac{1}{2} - i\frac{\sqrt{3}}{2}\right)^{32} + \left(\frac{3}{2} - i\frac{\sqrt{3}}{2}\right)^{32} = 2^{32} - 3^{16} - 1$$

30. The coefficient of x^8 is

$$\frac{1}{0!8!} + \frac{1}{2!6!} + \frac{1}{4!4!} + \frac{1}{6!2!} + \frac{1}{8!0!}$$

$$= \frac{1}{8!} [{}^8 C_0 + {}^8 C_2 + {}^8 C_4 + {}^8 C_6 + {}^8 C_8]$$

$$= \frac{2^7}{8!} = \frac{1}{315}$$

31. $23^{23} = 23 \cdot 23^{22} = 23(23^2)^{11}$

$$= 23(529)^{11} = 23(530-1)^{11}$$

$$= 23(53M-1), M \in N$$

$$= 23 \cdot 53M - 23$$

$$= 53(23M-1) + 30$$

Remainder is 30

32. $2^{2006} = 4(8)^{668} = 4(7+1)^{668}$ leaves remainder 4

$2006 = 7 \times 286 + 4$ leaves remainder

$\therefore 2^{2006} - 2006$ leaves remainder 0.

33. use multinomial theorem.

General term of given expansion is

$$\frac{6!}{a!b!c!} 1^a 2^{\frac{b}{2}} 3^{\frac{c}{3}} \text{ where } a+b+c=6 \text{ and}$$

a	b	c
-----	-----	-----

6	0	0
---	---	---

4	2	0
---	---	---

2	4	0
---	---	---

0	6	0
---	---	---

3	0	3
---	---	---

1	2	3
---	---	---

0	0	6
---	---	---

34. Required value is x^4 coefficient in

$$(1+x)^{404} \cdot {}^4C_0 - (1+x)^{303} \cdot {}^4C_1$$

$$+ (1+x)^{202} \cdot {}^4C_2 - (1+x)^{101} \cdot {}^4C_3 + {}^4C_4 - 1$$

Let $(1+x)^{101} = y$

$$\therefore {}^4C_0 y^4 - {}^4C_1 y^3$$

$$+ {}^4C_2 y^2 - {}^4C_3 y + {}^4C_4 - 1$$

$$= (y-1)^4 - 1$$

$$= [(1+x)^{101} - 1]^4 - 1$$

$$= [{}^{100}C_1 x + {}^{101}C_2 x^2 + \dots]^4 - 1$$

$$\therefore x^4 \text{ coefficient is } (101)^4$$

35. $\frac{2^{2003}}{17} = 8 \frac{16^{500}}{17}$

$$= \frac{8}{17} (1-17)^{500}$$

$$= \frac{8}{17} [1 - {}^{500}C_1 17 + {}^{500}C_2 17^2 + \dots]$$

Its fractional part is $\frac{8}{17}$

36. Put $x = \omega$ and ω^2 and add the two results



LEVEL-IV

STATEMENTS TYPE

1. (i) The no. of distinct terms in the expansion of $(x_1 + x_2 + \dots + x_n)^3$ is ${}^{n+2}C_3$

(ii) The no. of irrational terms in the expansion $(2^{1/5} + 3^{1/10})^{55}$ is 55

1) (i) is true (ii) is false 2) both (i) & (ii) are true
3) both (i) & (ii) are false 4) (i) is false (ii) is true

2. i. Three consecutive binomial coefficients can not be in G.P.

ii. Three consecutive binomial coefficients can not be in A.P.

Which of the above statement is correct?

1) both i and ii 2) neither i nor ii
3) Only i 4) Only ii

3. S_1 : If the coefficients of x^6 and x^7 in the

expansion of $\left(\frac{x}{4} + 3\right)^n$ are equal, then the number of divisors of n is 12.

S_2 : If the expansion of $\left(x^2 + \frac{2}{x}\right)^n$ for positive integer n has 13 th term independent of x . Then the sum of divisors of n is 39.

1) Only S_1 is true 2) Only S_2 is true
3) Both S_1 and S_2 are true
4) Neither S_1 nor S_2 is true

4. S_1 : The fourth term in the expansion of

$\left(2x + \frac{1}{x^2}\right)^9$ is equal to the second term in the

expansion of $(1+x^2)^{84}$ then the positive value

of x is $\frac{1}{2\sqrt{3}}$

S_2 : In the expansion of $\left(x^2 + \frac{a}{x^3}\right)^{10}$, the coefficients of x^5 and x^{15} are equal, then the positive value of a is 8

1) Only S_1 is true 2) Only S_2 is true
3) Both S_1 and S_2 are true
4) Neither S_1 nor S_2 is true

15. Assertion (A): Number of terms in the expansion of $(x + y + z)^5$ is 21.

Reason (R) : The number of terms in the expansion of $(x + y + z)^n$ is ${}^{(n+2)}C_2$

16. Assertion (A): In the expansion of $(x + x^{-2})^n$ the coefficient of eighth term and nineteenth term are equal then $n=25$

Reason (R): Middle term in the expansion of $(x + a)^n$ has greatest binomial coefficient where n is an even integer

17. Assertion (A): The expansion of $(1 + x)^n = C_0 + C_1x + C_2x^2 + \dots + C_nx^n$

Reason (R): If $x = -1$, then the above expansion is zero

18. Assertion (A): The coefficient of x^7 in $\left(\frac{x^2}{2} - \frac{2}{x}\right)^9$ is zero

Reason (R) : r in t_{r+1} that contain coefficient of x^7 is not positive integer

19. Assertion (A) : The coefficient of x^{-2} in the expansion of $\left(x^2 + \frac{1}{x}\right)^5$ is equal to 5C_4

Reason (R) : The value of r for the above (r in t_{r+1}) expansion is 3.

20. Assertion (A): The sum of the coefficients of the middle terms in the expansion of $(1 + x)^{2n-1}$ is equal to ${}^{2n}C_n$

Reason (R): To find the sum of the coefficients of the two middle terms in the expansion $(1 + x)^{2n-1}$ the Value of n is any natural number.

21. Assertion (A): In $(1 + x)^n$ sum of coefficients of even powers of x is not equal to the sum of coefficients of odd powers of x .

Reason (R): The value $(1 + x)^n$ for $x = -1$ is zero

22. Assertion (A) : Greatest binomial coefficient in the expansion of $(1 + 5x)^8$ is ${}^8C_45^4$

Reason (R) : Greatest coefficient in the expansion of $(1 + x)^{2n}$ is the middle term.

MATRIX MATCHING TYPE

23. Let $R = (5\sqrt{5} + 11)^{2n+1}$ and $F = R - [R]$ where $[]$ denotes greatest integer less than or equal to R , and if $f = (5\sqrt{5} - 11)^{2n+1}$

List - I

List - II

A) $[R]$

1) 0

B) RF

2) 4^{2n+1}

C) $f-F$

3) odd integer

D) $R-f$

4) even integer

5) 4^{2n-1}

The correct match is

	A	B	C	D		A	B	C	D
1)	2	3	1	5	2)	1	5	2	3
3)	3	1	2	4	4)	4	2	1	4

24. Observe the following lists :

List-I

List-II

(a) The number of terms in the

(i) 12

expansion of $(x + y - z)^6$

(b) The number of terms which depend on x in the expansion of

(ii) 20

$$\left(x^2 + 2 + \frac{1}{x^2}\right)^6$$

(c) The sum of the coefficients in the expansion of $(1 - x + 2x^2)^4$

(iii) 28

(d) The greatest coefficient in $(1 + x)^6$ is

(iv) 16

Find the correct match for List I from List II

1) $a \rightarrow (iii)$ $b \rightarrow (i)$ $c \rightarrow (ii)$ $d \rightarrow (iv)$

2) $a \rightarrow (iv)$ $b \rightarrow (i)$ $c \rightarrow (ii)$ $d \rightarrow (iii)$

3) $a \rightarrow (iii)$ $b \rightarrow (i)$ $c \rightarrow (iv)$ $d \rightarrow (ii)$

4) $a \rightarrow (iv)$ $b \rightarrow (i)$ $c \rightarrow (iii)$ $d \rightarrow (ii)$

LEVEL-IV - KEY

- 1) 1 2) 3 3) 3 4) 4 5) 1 6) 3 7) 1
 8) 2 9) 2 10) 2 11) 4 12) 1 13) 1 14) 4
 15) 1 16) 2 17) 2 18) 1 19) 3 20) 2 21) 4
 22) 4 23) 4 24) 3

LEVEL-IV - HINTS

- 1) Formula
2) False
- Standard Results
- Coeff of T_6 = Coeff of T_7

$${}^nC_{n-6} \left(\frac{1}{4}\right)^6 \cdot (3)^{n-6} = {}^nC_{n-7} \left(\frac{1}{4}\right)^7 \cdot (3)^{n-7}$$

$$\rightarrow n = 90$$

$$s_2) 13 = \frac{2n}{3} \Rightarrow n = \frac{39}{2} \text{ not possible}$$
- S_1) Find 4th and second term
 S_2) Find coeff. x^5 and x^{15}
- 1) Standard result 2) Wrong
- A) Standard result
 B) Coefficient of x^n in $(1+x)^{2n} = {}^{2n}C_n$

$$C) \frac{(2n)!}{(n!)^2}$$
- Use the formula $r = \frac{np}{p+q}$ in all cases and arranging ascending order
- A) $r = \frac{5}{1/2+2}$
 $T_3 = 405$ find n
 B) $T_3 = 1000$ frame the equation and verify the option
 C) $(n-2r)^2 = n+2$

- A) $T_{r+1} = \frac{(r+1)(r+2)}{2!} x^r$
 B) $T_{r+1} = (r+1)x^r$ C) $T_{r+1} = (-1)^r x^r$
 D) Direct expansion
- $n > 6, \left(\frac{n}{2}\right)^n > n! > \left(\frac{n}{3}\right)^n$
- Number of terms 52
- $P = 2^n \{2^{3n} - (7n+1)\}$

$$= 2^n \{(1+7)^n - (1+7n)\}$$

$$= (14)^2 \cdot 2^{n-2} \{n_{C_2} + n_{C_3} 7 + \dots\} ; (n \geq 2)$$
- Put $x=1$
- $m = 6 \Rightarrow T_3 = 240$ $30 {}_{C_1} x^3 = 240 \Rightarrow x = 2$
- Use the formula ${}^{n+r-1}C_{r-1}$
- $n_{C_7} = n_{C_{18}} \Rightarrow n = 25$
- Standard results and they are independent relations
- $\frac{18-k}{3} = 7$
 A) $21-18 = -k$ not possible $\Rightarrow$ coefficient = 0
 R) r is a fraction
- $r = \frac{10+2}{3} = 4$
 A) $T_5 = {}^5C_4$
 R) $r = 4$
- A) $T_n + T_{n+1} = {}^{2n}C_n$
 R) Middle term has to be an integer
- A) Sum of the coefficients of even powers of x = sum of the coefficients of odd powers
 R) Put $x = -1$
- Greatest Binomial coefficients is 8C_4
- Verify the options
- Simplify

